



ashka singh <sgsinghashka@gmail.com>

TA || Ashka || TM- B141

ashka singh <sgsinghashka@gmail.com>
To: dhiraj.aiworkspace@gmail.com

11 April 2026 at 16:28

Dear Sir,

Regarding the topics of the Basic UI Elements – title, header, text, Streamlit Buttons and Callbacks, Index Basics, INNER JOIN, Introduction to Views, LEFT JOIN, Streamlit Selection Widgets – Selectbox, Multiselect, Radio and Streamlit Input Widgets – text_input, number_input, kindly find the requested details below.

Date: 4th April 2026

Topic: Basic UI Elements – title, header, text

Technology: SQL

1. What is the difference between st.title() and st.header()?

Ans: The primary difference between st.title() and st.header() in Streamlit is their hierarchy and visual size. They are both wrappers around Markdown that allow us to display text in a structured way.

2. When should you use st.text() instead of st.write()?

Ans: In Streamlit, we should use st.text() when we need to display raw, preformatted text exactly as it is, without any styling or Markdown parsing. In contrast, st.write() is a multipurpose "Swiss Army knife" that attempts to intelligently format its input.

3. Can we use multiple titles in one app?

Ans: Yes, you can manage multiple titles for a single application in SQL.

4. How do UI elements improve user experience?

Ans: UI elements improve user experience by making digital interfaces intuitive, efficient, and engaging.

5. Which function is best for large headings?

Ans: SELECT TOP 100 * FROM table_name;

Date: 4th April 2026

Topic: Streamlit Buttons and Callbacks

Technology: SQL

1. What is the purpose of st.button() in Streamlit?

Ans: The primary purpose of st.button() in Streamlit is to create a clickable button widget that allows users to trigger specific actions or execute code blocks only when clicked.

2. What is a callback function?

Ans: A callback function is a function passed into another function as an argument, which is then executed (or "called back") inside the outer function to complete a specific routine, action, or asynchronous operation.

3. How does session_state help with buttons?

Ans: `st.session_state` in Streamlit helps with buttons by persisting data across the script's reruns, which normally reset whenever a button is clicked. Because Streamlit reruns the entire script from top to bottom on interaction, session state allows us to store the button's state ("pressed" vs "not pressed") or other data, preventing it from disappearing.

4. What happens if two buttons have the same callback?

Ans: When two buttons share the same callback, clicking either button triggers the exact same function.

5. How can you prevent a function from running repeatedly in Streamlit?

Ans: To prevent a specific function from running repeatedly during these reruns, we can use built-in tools like caching, session state, forms, and fragments.

Date: 4th April 2026

Topic: Index Basics Technology: SQL

1. What is an index in SQL?

Ans: Indexes in SQL are special database structures that speed up data retrieval by allowing quick access to records instead of scanning the entire table. They act like a lookup system and play an important role in improving query performance and database efficiency.

- It speed up queries (SELECT, JOIN, WHERE, ORDER BY).
- Reduce disk I/O and improve efficiency in large tables.
- Ensure data integrity with unique indexes.

2. How does an index improve performance?

Ans: Indexes improve SQL query performance by reducing the amount of data the database must scan to find results. Without an index, the database performs a full table scan, which checks every row in a table to satisfy a query. This is inefficient, especially for large tables.

3. Can index slow down operations?

Ans: Yes, database indexes can significantly slow down certain operations, specifically data modification, even though they are designed to speed up data retrieval (SELECT queries).

4. What is a composite index?

Ans: A composite index (also known as a multi-column index) is a database index created on two or more columns of a table.

5. When should we avoid using index?

Ans: Avoid using database indexes on small tables, columns with low cardinality (few unique values), or tables with frequent write operations (INSERT, UPDATE, DELETE).

Date: 4th April 2026

Topic: INNER JOIN

Technology: SQL

1. What is INNER JOIN?

Ans: An INNER JOIN in SQL selects records that have matching values in both tables, returning only the rows where the join condition is met.

2. When do we use INNER JOIN?

Ans: Use an INNER JOIN in SQL when we need to combine rows from two or more tables based on a related column, returning *only* the records that have matching values in both tables.

3. What happens if no matching record exists?

Ans; When no matching record exists in a SQL join, the database typically returns NULL values for all columns from the table that lacks a match. Specifically, with a LEFT JOIN, all rows from the left table are returned, and if no match exists on the right, the right-side columns display as NULL.

4. Difference between INNER JOIN and LEFT JOIN?

Ans:

Feature	INNER JOIN	LEFT (OUTER) JOIN
Result Set	Includes only rows with matching values in both tables.	Includes all rows from the left table and matched rows from the right.
Unmatched Rows	Discarded entirely.	Preserved from the left table; right-side columns are filled with NULL.
Mathematical Concept	Intersection ($A \cap B$).	Left table (A) + Intersection ($A \cap B$).
Common Use Case	When you only want "complete" records (e.g., customers who have made orders).	When you want to see everything from one list, even if details are missing (e.g., all students, regardless of enrollment).

5. What column is used for INNER JOIN?

Ans: An INNER JOIN uses a related column (often a foreign key in one table and the primary key in another) to match rows between two tables. The columns must share a common value or relationship, defined in the ON clause (e.g., TableA.id = TableB.foreign_id) to return only rows with matches in both tables.

Date: 4th April 2026

Topic: Introduction to Views

Technology: SQL

1. What is a view in SQL?

Ans: A view in SQL is a virtual table based on the result-set of an SQL statement.

2. Difference between table and view?

Ans: In a relational database, the fundamental difference is that a Table is a physical storage container for data, while a View is a "virtual table" that is actually a stored SQL query.

3. Why do we use views?

Ans: In SQL, a view is a virtual table based on the result-set of an SQL statement. Unlike a standard table, it does not store data physically but instead pulls data from underlying "base" tables dynamically when queried.

4. Can views improve security?

Ans: Yes, SQL views significantly improve security by restricting user access to sensitive data, limiting visibility to specific rows or columns, and enforcing read-only permissions. By granting users access to a view rather than the underlying base table, organizations can hide sensitive information, reduce data exposure risk, and enforce data integrity, as detailed by [OneUptime](#).

5. How do you delete a view?

Ans: In database management, "delete view" typically refers to permanently removing the view's definition from the database. This is performed using the DROP VIEW statement.

Date: 4th April 2026

Topic: LEFT JOIN

Technology: SQL

1. What is LEFT JOIN?

Ans: A LEFT JOIN (or LEFT OUTER JOIN) in SQL returns all records from the left table and matched records from the right table.

3. What does NULL indicate in LEFT JOIN result?

Ans: In a LEFT JOIN (or LEFT OUTER JOIN), a NULL result typically indicates that no matching record was found in the right table for a row existing in the left table.

4. When should we use LEFT JOIN?

Ans: Use a LEFT JOIN (or LEFT OUTER JOIN) when we want to retrieve all records from the left table and only the matching records from the right table.

5. Which table is called left table?

Ans: In SQL, the left table is the first table mentioned in the FROM clause or the table that appears immediately to the left of the [LEFT JOIN](#) keyword.

Date: 4th April 2026

Topic: Streamlit Selection Widgets – Selectbox, Multiselect, Radio

Technology: SQL

1. What is the difference between selectbox() and radio()?

Ans: [selectbox\(\)](#) (dropdown) and [radio\(\)](#) (radio buttons) both allow single-choice selection from a list but differ in visibility and space usage. Selectbox hides options until clicked, making it ideal for >6 -10 choices or limited space, while radio buttons reveal all choices, offering better visibility for 5 < Options.

2. When should multiselect() be used?

Ans: The multiselect() component should be used when users need to select one or more items from a long, predefined list while minimizing screen space, such as for filters, tags, or assigning multiple categories.

3. What type of data does `multiselect()` return?

Ans: `multiselect()` (specifically `st.multiselect` in Streamlit) returns a list of strings containing the options selected by the user.

4. Can `selectbox()` have a default value?

Ans: Yes, `st.selectbox()` in Streamlit can have a default value. We can set it by using the `index` parameter, which accepts the integer index (starting from 0) of the option we want to be selected by default.

5. How do selection widgets improve user experience?

Ans: Selection widgets improve user experience by streamlining data input, reducing cognitive load, and enabling efficient interaction with digital interfaces.

Date: 4th April 2026

Topic: Streamlit Input Widgets – `text_input`, `number_input`

Technology: SQL

1. What is the difference between `text_input()` and `number_input()`?

Ans: The core difference between `st.text_input()` and `st.number_input()` is the expected input type and widget functionality. `text_input()` accepts any string, allowing text and symbols, while `number_input()` restricts input to numerical values (integers or floats), often providing plus/minus arrows for adjustment.

2. How can you limit the values in `number_input()`?

Ans: In Streamlit, we can limit the values in a `st.number_input()` widget using the `min_value`, `max_value`, and `step` parameters within the function call.

3. What data type does `text_input()` return?

Ans: The `st.text_input()` function in Streamlit returns a string (`str`) data type. When used in SQL queries, this input is treated as character data (such as `VARCHAR`, `TEXT`, or `NVARCHAR`) and must be properly handled to prevent SQL injection, usually by wrapping it in quotes within the SQL string or using parameterized queries.

4. What happens if `text_input()` is empty?

Ans: When a `text_input()` field is empty in SQL-related contexts (like applications or forms), it typically passes an empty string (`""`) or a `NULL` value to the database, depending on the framework, rather than omitting the parameter entirely. This usually results in storing an empty string, allowing a search for blank entries, or potentially causing errors if the column is defined as `NOT NULL`.

5. How do input widgets help in building interactive apps?

Ans: Input widgets (sliders, text inputs, buttons, checkboxes) build interactive apps by enabling real-time user input, allowing users to manipulate data, trigger actions, and customize views instantly without refreshing the page.

Regards

Ashka Singh

----- Forwarded message -----

From: **ashka singh** <sgsinghashka@gmail.com>

Date: Sat, 4 Apr 2026 at 17:59

Subject: Fwd: TA || Ashka || TM- B141

To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topics of Aggregate Functions (SUM, COUNT, AVG), Comparison Operators in SQL, Constraints in SQL, Creating a Database, SQL, Creating a Table, Data Types in SQL, Deleting Data, Foreign Key, GROUP BY, HAVING Clause, Index Basics, INNER JOIN, Inserting Data, Introduction to SQL, Introduction to Views, Joins Overview, LEFT JOIN, Limiting Results, Logical Operators (AND, OR, NOT), NULL Values, Primary Key, Relational Database Concept, Selecting Data, Sorting Data using ORDER BY, Subqueries Basics, Tables, Rows, and Columns, Tables, Rows, and Columns, Types of Databases, Updating Data, What is Database and WHERE Clause. Kindly find the requested details below.

Date: 28th March 2026

Topic: Aggregate Functions (SUM, COUNT, AVG)

Technology: SQL

1. What are aggregate functions?

Ans: Aggregate functions are built-in operations that perform a calculation on a set of values across multiple rows and return a single summarized value.

2. Difference between COUNT(*) and COUNT(column)?

Ans:

Feature	COUNT(*)	COUNT(column)
What is counted?	All rows in the table/group.	Only rows with non-NULL values in that column.
NULL Handling	Includes NULLs.	Excludes NULLs.
Duplicates	Includes duplicates.	Includes duplicates (unless DISTINCT is used).
Performance	Highly optimized; often uses the smallest available index.	May require a scan of the specific column to check for NULLs.

3. Can SUM work on text?

Ans: No, the standard SQL SUM() function does not work directly on text (e.g., VARCHAR, CHAR, or TEXT data types). It is strictly an aggregate function designed for numeric data types such as INT, DECIMAL, FLOAT, and REAL.

4. What happens if column has NULL values?

Ans: In SQL, a NULL value represents missing, unknown, or inapplicable data. It is a special marker, not a value like zero (0) or an empty string ("").

5. How are aggregate functions used with GROUP BY?

Ans: Aggregate functions are used with the GROUP BY clause to summarize data into meaningful categories.

Topic: Comparison Operators in SQL
Technology: SQL

1. What are comparison operators in SQL?

Ans: SQL Comparison Operators are used to compare two values and filter the result set based on a specific condition.

- Used in the WHERE clause to filter records based on conditions
- Help check for equality, range, or inequality
- Make SQL queries powerful and precise
- Examples: =, >, <, >=, <=, <>

Equal to	=
Not equal to	<> or !=
Greater than	>
Less than	<
Greater than or equal to	>=
Less than or equal to	<=

2. Difference between = and <> ?

Ans: Comparison Table

Feature	=	<>
Logic	Checks for equality	Checks for inequality
Standard	ISO/ANSI Standard	ISO/ANSI Standard
Alternative	N/A	!= (Non-standard)
Example	id = 5	id <> 5

3. Where are comparison operators used?

Ans: In SQL, comparison operators (also known as relational operators) are primarily used in conditional clauses to filter and refine data. They compare two expressions and return a boolean result: TRUE, FALSE, or UNKNOWN (if a NULL value is involved).

4. What is the meaning of >= ?

Ans: In SQL, the >= symbol is a comparison operator that means "Greater Than or Equal To".

5. Can comparison operators be used with text values?

Ans: Yes, comparison operators can be used with text values in SQL. When used with strings.

Topic: Constraints in SQL
Technology: SQL

1. What are constraints in SQL?

Ans: Constraints are rules applied to table columns to ensure data integrity, accuracy, and reliability.

2. What is the difference between UNIQUE and PRIMARY KEY?

Ans:

Feature	Primary Key	Unique Key
Number	Only one per table	Multiple per table
NULL values	Not allowed	Allowed (usually one)
Index	Clustered	Non-clustered
Usage	Identifies a row	Ensures uniqueness

3. Why is NOT NULL important?

Ans: The NOT NULL constraint is a fundamental SQL tool used to ensure that a column always contains a value. By default, SQL columns allow NULL values, which represent missing, unknown, or inapplicable data.

4. How does CHECK constraint work?

Ans: A CHECK constraint is a database rule used to limit the range of values that can be stored in one or more columns. It ensures that data inserted or updated in a table adheres to a specific condition, maintaining data integrity and enforcing business logic.

5. Can constraints be added after table creation?

Ans: Yes, constraints can be added after a table has been created using the ALTER TABLE statement.

Topic: **Creating a Database**

Technology: SQL

1. What is the command to create a database?

Ans: The standard SQL command to create a new database is CREATE DATABASE, followed by the name we wish to give the database.

SQL

CREATE DATABASE database_name;

2. Why do we use USE command?

Ans: The USE command selects a specific database from multiple available databases on a server and sets it as the default context for all subsequent queries.

3. Can we create tables without creating a database?

Ans: In strictly technical SQL terms, we cannot create a table without it belonging to a database. Every table must reside within a database or schema to be stored and accessed.

4. What is the difference between database and table?

Ans: A database is a high-level container used to store and organize information, while a table is a specific structure within that database where the actual data is stored in rows and columns.

5. How do you delete a database?

Ans: To delete a database in SQL, the primary command used is DROP DATABASE.

Topic: **Creating a Table**

Technology: SQL

1. What is the syntax to create a table?

Ans: The standard syntax for the CREATE TABLE statement in SQL is as follows:

```
CREATE TABLE table_name (  
    column1 datatype constraints,  
    column2 datatype constraints,  
    column3 datatype constraints,  
    ....  
);
```

2. Why do we define data types while creating a table?

Ans: Defining data types during table creation is essential for establishing a structured, efficient, and reliable database.

3. What happens if data type is not correct?

Ans: Using an incorrect data type in SQL can lead to various issues ranging from immediate operation failures to long-term performance degradation.

4. Can a table exist without columns?

Ans: In most SQL database systems, a table cannot exist without at least one column.

5. What is the difference between table and database?

Ans: In SQL, the primary difference is that a database is a container for holding data, while a table is the specific structure within that container where data is organized and stored.

Topic: **Data Types in SQL**

Technology: SQL

1. What are SQL data types?

Ans: SQL data types specify the kind of data a table column can hold, such as numbers, text, or dates. They are critical for ensuring data integrity, optimizing storage efficiency, and improving query performance.

2. What is the difference between CHAR and VARCHAR?

Ans: The main difference between CHAR and VARCHAR is that CHAR is a fixed-length data type while VARCHAR is a variable-length data type. CHAR always reserves the maximum specified storage space, padding shorter strings with spaces, whereas VARCHAR uses only the space required for the actual data plus a small overhead for length information.

3. Why is INT used instead of VARCHAR for numbers?

Ans: In database design, choosing INT (Integer) over VARCHAR (Variable Character) for numeric data is standard practice due to performance, storage efficiency, and data integrity.

4. What happens if wrong data type is used?

Ans: Using the wrong data type typically leads to [program errors](#), data corruption, or unexpected application behavior.

5. Can we store numbers in VARCHAR?

Ans: Yes, we can store numbers in a VARCHAR column, as it is designed to hold alphanumeric strings, including digits. However, this is generally considered a poor practice unless the data is an identifier rather than a mathematical value.

Topic: **Deleting Data**

Technology: SQL

1. What is DELETE command?

Ans: The DELETE command is a Data Manipulation Language (DML) statement used to remove one or more existing records (rows) from a table.

2. Difference between DELETE and DROP?

Ans: Use [DELETE](#): When we want to remove specific records—like deleting a single user or all orders from last year—while keeping the table ready for new data.

- Use [DROP](#): When we no longer need the table at all, such as removing a temporary table used for a one-time report.

3. What happens if WHERE is missing? Ans: The WHERE clause is an optional component used to filter records. If it is omitted, the statement will apply to all rows in the specified table.

4. Can we recover deleted data?

Ans: Yes, we can recover deleted data in SQL, though the method depends on whether we have a backup and the specific database settings in place at the time of deletion.

5. How do you delete all rows but keep table?

Ans: To delete all rows while keeping the table structure, use either the TRUNCATE TABLE or DELETE statement.

Topic: **Foreign Key**

Technology: SQL

1. What is a foreign key?

Ans: A foreign key is a column (or a set of columns) in one table that refers to the primary key (or a unique key) in another table.

2. How is foreign key different from primary key?

Ans:

PRIMARY KEY	FOREIGN KEY
A primary key is used to ensure data in the specific column is unique.	A foreign key is a column or group of columns in a relational database table that provides a link between data in two tables.
It uniquely identifies a record in the relational database table.	It refers to the field in a table which is the primary key of another table.
Only one primary key is allowed in a table.	Whereas more than one foreign key is allowed in a table.
It is a combination of UNIQUE and Not Null constraints.	It can contain duplicate values and a table in a relational database.

It does not allow <u>NULL values</u> .	It can also contain NULL values.
Its value cannot be deleted from the parent table.	Its value can be deleted from the child table.
The constraint can be implicitly defined on the temporary tables.	The constraint cannot be defined on the local or global temporary tables.

3. Can foreign key contain NULL values?

Ans: Yes, a Foreign Key can contain NULL values in SQL.

4. What happens if foreign key value does not exist in parent table?

Ans: When a foreign key value does not exist in the parent (referenced) table, the following typically occurs in a standard SQL database:

- Operation Failure: An attempt to INSERT or UPDATE a record in the child table with a non-existent parent value will be rejected.
- Constraint Violation Error: The database will throw an error message indicating a Foreign Key Constraint Violation.

For example:

- SQL Server: "The INSERT statement conflicted with the FOREIGN KEY constraint..."

5. Why do we use foreign keys?

Ans: A foreign key is a column (or group of columns) in one table that points to a primary key in another table. We use them primarily to establish and enforce relationships between data sets.

Topic: **GROUP BY**
Technology: SQL

1. What is GROUP BY?

Ans: The GROUP BY clause is used to arrange rows with identical data into summary groups. It is almost always used in combination with aggregate functions to perform calculations on each group.

2. Why is GROUP BY used with aggregate functions?

Ans: The GROUP BY clause is used with aggregate functions (like SUM, COUNT, AVG, MAX, and MIN) to partition a result set into specific subsets of data before performing a calculation.

3. What happens if GROUP BY is missing?

Ans: When a GROUP BY clause is missing in an SQL query that uses aggregate functions, the behavior depends on whether other non-aggregated columns are included in the SELECT statement.

4. Can we use WHERE with GROUP BY?

Ans: Yes, we can use the WHERE clause with GROUP BY in a single SQL query.

When used together, they serve different purposes at different stages of the query's execution:

- WHERE: Filters individual rows *before* they are grouped.
- GROUP BY: Aggregates the remaining rows into summary groups.

5. What is difference between GROUP BY and ORDER BY?

Ans:

Feature	GROUP BY	ORDER BY
Purpose	Groups rows with the same values into summary rows.	Sorts the result set in ascending or descending order.
Common Use	Almost always used with aggregate functions like SUM(), AVG(), or COUNT().	Used to organize data numerically, alphabetically, or by date.
Output Rows	Reduces the number of rows (returns one row per group).	Keeps all rows but changes their sequence.
Execution	Executed before ORDER BY.	Executed after GROUP BY.
Filtering	Uses the HAVING clause to filter grouped results.	Uses the WHERE clause to filter individual rows before sorting.

Topic: **HAVING Clause**

Technology: SQL

1. What is HAVING clause?

Ans: The HAVING clause is used to filter the results of a query after they have been grouped and aggregated.

2. Difference between WHERE and HAVING?

Ans:

Feature	WHERE Clause	HAVING Clause
Primary Purpose	Filters individual rows.	Filters groups of rows.

Execution Order	Applied before data is grouped.	Applied after data is grouped.
Aggregate Functions	Cannot be used with aggregates (e.g., SUM, AVG).	Can be used with aggregate functions.
Usage Statements	Used with SELECT, UPDATE, and DELETE.	Typically used only with SELECT.
Requirement	Can be used without a GROUP BY clause.	Generally requires a GROUP BY clause.

3. Can HAVING work without GROUP BY?

Ans: Yes, the HAVING clause can work without a GROUP BY clause in SQL. When used this way, the entire result set is treated as a single group.

4. Why do we use HAVING with aggregate functions?

Ans: The HAVING clause is used to filter result sets based on aggregate functions because the WHERE clause cannot be used for this purpose.

5. When should HAVING be used?

Ans: The HAVING clause is used to filter results after data has been grouped and aggregated. It was specifically introduced because the WHERE clause cannot be used with aggregate functions.

Topic: **Index Basics**

Technology: SQL

1. What is an index in SQL?

Ans: An index is a performance-optimization structure that speeds up data retrieval from a database table, much like the index at the back of a textbook helps you find a specific topic without reading every page.

2. How does an index improve performance?

Ans: An index improves performance by acting as a lookup guide that allows the database engine to locate specific data without scanning every row in a table. This process, known as an Index Seek, is significantly faster than a "Full Table Scan," especially as data volume grows.

3. Can index slow down operations?

Ans: Yes, indexes can slow down database operations.

4. What is a composite index?

Ans: A composite index (also known as a multi-column index) is a database index created on two or more columns of a single table. It acts as a specialized lookup tool that allows the database engine to quickly locate rows based on a combination of column values rather than searching each column separately.

5. When should we avoid using index?

Ans: Indexes should be avoided in situations where the overhead of index maintenance, storage, and potential performance degradation of write operations outweighs the benefits of faster read (SELECT) operations.

Topic: INNER JOIN

Technology: SQL

1. What is INNER JOIN?

Ans: An INNER JOIN is a command used to combine rows from two or more tables based on a related column between them.

2. When do we use INNER JOIN?

Ans: An INNER JOIN is used to combine rows from two or more tables based on a related column between them.

3. What happens if no matching record exists?

Ans: When no matching record exists in SQL, the result depends on the join type: a [LEFT JOIN](#) returns NULL values for the right table's columns, while an [INNER JOIN](#) excludes the row entirely. For identification, NOT EXISTS or [WHERE ... IS NULL](#) queries are used to specifically return unmatched rows.

4. Difference between INNER JOIN and LEFT JOIN?

Ans:

Feature	INNER JOIN	LEFT JOIN
Match Requirement	Must exist in both tables	Only required in the left table
Unmatched Rows	Discarded	Preserved from the left table
Missing Data	Not shown	Shown as NULL for the right table
Common Use Case	Finding commonalities	Finding orphans or reporting on all items

5. What column is used for INNER JOIN?

Ans: An INNER JOIN uses common columns (foreign keys) that exist in both tables to match and return rows where values exist in both, typically linking a primary key from one table to a foreign key in another. The specific column names must match, such as TableA.ID = TableB.TableA_ID in the ON clause.

Topic: Inserting Data

Technology: SQL

1. What is INSERT command?

Ans: The INSERT command (specifically INSERT INTO) is a fundamental SQL statement used to add new records (rows) to a database table.

2. How do you insert multiple records?

Ans: INSERT INTO table_name (column1, column2, column3, ...)

VALUES

(value1, value2, value3, ...),

(value4, value5, value6, ...),

(value7, value8, value9, ...);

3. What happens if primary key is duplicate?

Ans: In SQL, a Primary Key must contain unique values to identify each row in a table. If we try to insert or update a row with a primary key that already exists, the database engine will reject the operation and throw an error to maintain [data integrity](#).

4. Can we insert data without column names?

Ans: Yes, we can insert data into a table without specifying column names.

5. What error occurs if data type is wrong?

Ans: When a data type in an SQL statement is incorrect or incompatible with the target column, it typically triggers a Data Type Mismatch error.

Topic: **Introduction to SQL**

Technology: SQL

1. What is SQL?

Ans: SQL (Structured Query Language) is the standard [programming language](#) used to interact with relational database management systems (RDBMS).

2. Why is SQL used?

Ans: [Structured Query Language \(SQL\)](#) is a standard programming language used to manage, manipulate, and retrieve data stored in relational databases. It allows users to create, update, delete, and query data across tables, making it essential for backend development, data analysis, and database administration. Common uses include running, searching, and updating data records.

3. Difference between database and table?

Ans:

Feature	Database	Table
Definition	An organized collection of interrelated data and objects.	A two-dimensional structure of rows and columns within a database.
Scope	A high-level container that can hold thousands of tables.	A lower-level component that stores specific data records.
Contents	Tables, views, indexes, stored procedures, and security rules.	Data entries (rows) and category labels (columns).
Primary Goal	To manage, secure, and update the entire data ecosystem.	To store data about a specific entity or subject.

4. What are types of SQL commands?

Ans: There are four types of commands: preparatory commands, commands of execution, combined commands, and supplementary commands.

5. Where is SQL used in real life?

Ans: SQL (Structured Query Language) is the standard language used to communicate with relational databases. In real life, it acts as the "engine" behind nearly every digital interaction involving stored data—from scrolling through social media to checking a bank balance.

Topic:**Introduction to Views**
Technology: SQL

1. What is a view in SQL?

Ans: In SQL, a view is a virtual table that represents the result-set of a stored [SELECT query](#).

2. Difference between table and view?

Ans: A table is a physical storage structure that holds actual data, while a view is a virtual object that represents data from one or more tables.

3. Why do we use views?

Ans: Views in SQL are virtual tables created from queries, used primarily to simplify complex joins, enhance security by restricting data access, and promote code reusability.

4. Can views improve security?

Ans: Yes, views (virtual tables in SQL databases) can significantly improve security by acting as a security layer that restricts, masks, and simplifies data access.

5. How do you delete a view?

Ans: To permanently remove a view from a database, the SQL DROP VIEW statement is used. This command removes only the view definition and associated permissions, leaving the original base tables and their data intact.

Topic:**Joins Overview**
Technology: SQL

1. What is a join in SQL?

Ans: A JOIN in SQL is a clause used to combine rows from two or more tables based on a related column between them.

2. Difference between INNER JOIN and LEFT JOIN?

Ans:

Feature	INNER JOIN	LEFT JOIN
Result Set	Includes only matching rows from both tables.	Includes all rows from the left table plus matching rows from the right.
Unmatched Data	Excludes unmatched rows entirely.	Preserves left table rows; right table columns show NULL for missing matches.
Standard Usage	Use when you only need data that exists in both sources (e.g., customers who have placed orders).	Use when you need a complete list from one table, even if some have no related data (e.g., all customers, even those with no orders).

Alternative Names	JOIN (INNER is often the default).	LEFT OUTER JOIN (the terms are functionally identical).
-------------------	------------------------------------	---

3. What happens if there is no matching data?

Ans: Inner Join: If there is no match between the joined tables, the entire row is excluded from the result set.

- Outer Joins (LEFT, RIGHT, FULL): These preserve rows even without a match. The columns from the table lacking matching data will be filled with NULL.

4. Can we join more than two tables?

Ans: Yes, we can join more than two tables in a single SQL query to combine related data from multiple sources. By chaining multiple JOIN statements (such as [INNER JOIN](#) or [LEFT JOIN](#)), you can link three or more tables together.

5. What column is used for joining tables?

Ans: Tables are joined using common columns that contain related data. While any column with matching data types can technically be used, the process typically relies on specific relational keys:

- Primary Key: A unique identifier for each row in its own table (e.g., CustomerID in a Customers table).
- Foreign Key: A column in a different table that references a primary key to create a link (e.g., CustomerID in an Orders table).

Topic: **LEFT JOIN**

Technology: SQL

1. What is LEFT JOIN?

Ans: A [LEFT JOIN](#) (or LEFT OUTER JOIN) in SQL returns all records from the left table and the matching records from the right table.

2. Difference between LEFT JOIN and INNER JOIN?

Ans:

Feature	INNER JOIN	LEFT JOIN
Logic	Returns only rows with matches in both tables.	Returns all rows from the left table, plus matches from the right.
Unmatched Data	Excludes non-matching rows entirely.	Keeps unmatched left rows and fills right columns with NULL.
Result Set Size	Usually smaller; only the "intersection".	Usually larger; at least as many rows as the left table.

Default Behavior	Often the default type when just using JOIN.	Must be explicitly defined as LEFT JOIN or LEFT OUTER JOIN.
------------------	--	---

3. What does NULL indicate in LEFT JOIN result?

Ans: In a LEFT JOIN result, a NULL value in a column from the right-hand table indicates that no matching record was found in that table for a specific row in the left-hand table.

4. When should we use LEFT JOIN?

Ans: Use a LEFT JOIN (or LEFT OUTER JOIN) when we want to retrieve all records from the left table and only the matching records from the right table.

5. Which table is called left table?

Ans: In SQL, the left table is the table that appears immediately after the FROM clause (or the preceding table in a series of joins). It is the table that is listed first in the SQL query before the LEFT JOIN keyword.

Topic: Limiting Results

Technology: SQL

1. What is LIMIT in SQL?

Ans: In SQL, the LIMIT clause is used to restrict the number of records returned by a query.

2. Why do we use LIMIT with ORDER BY?

Ans: We use LIMIT with ORDER BY in SQL to retrieve a specific, ranked subset of data (like the "top 10" or "bottom 5") efficiently, rather than returning all rows.

3. What is OFFSET?

Ans: The OFFSET clause in SQL is used to skip a specific number of rows from the beginning of a result set before returning any records.

4. What is SQL Server alternative for LIMIT?

Ans: In SQL Server, the direct equivalents to the MySQL/PostgreSQL LIMIT clause are the TOP keyword and the OFFSET-FETCH clause.

1. SELECT TOP Clause

The TOP clause is the most common equivalent used to limit the total number of rows returned by a query. It is placed immediately after the SELECT keyword.

- Basic Syntax: SELECT TOP (number) column_names FROM table_name;

Example: To get the first 10 rows from a table **Sql** SELECT TOP 10* FROM Employees ORDER BY HireDate DESC;

5. Can LIMIT improve performance?

Ans: Yes, using the LIMIT clause can significantly improve SQL performance by reducing the workload on the database engine, network, and application.

Date: 28th March 2026

Topic: **Logical Operators (AND, OR, NOT)**

Technology: SQL

1. What is the difference between AND and OR?

Ans:

Feature	AND Operator	OR Operator
Logic	Returns a record only if all conditions are true.	Returns a record if any condition is true.
Filtering	Narrows down results (more restrictive).	Broadens results (more flexible).
Precedence	Evaluated before OR when used in the same statement.	Evaluated after AND unless parentheses are used.

2. How does NOT operator work?

Ans: The NOT operator is a logical operator used to negate a boolean condition. It reverses the truth value of the expression that follows it: if a condition is TRUE, it becomes FALSE, and vice versa.

3. Can we combine AND and OR in one query?

Ans: Yes, we can combine AND and OR operators in a single SQL query to create complex filtering conditions.

4. What happens if logical operators are not used correctly?

Ans: When logical operators (AND, OR, NOT) are not used correctly in SQL, the database will often execute the query without a syntax error, but the results will be incorrect or the system's performance will suffer.

5. Give a real example of logical operators.

Ans: AND Operator (Multiple Requirements):

To find specific items, such as Amazon product reviews for a specific product that also have a high star rating.

```
SELECT * FROM reviews
```

```
WHERE product_id = 50001 AND stars > 3;
```

Topic: **NULL Values**

Technology: SQL

1. What is NULL in SQL?

Ans: In SQL, NULL is a special marker used to represent missing, unknown, or inapplicable data in a database field.

2. Difference between NULL and 0?

Ans:

Feature	0 (Zero)	NULL
---------	----------	------

Meaning	A specific, known numeric value representing "none" or an empty quantity.	An unknown, missing, or non-applicable value representing the absence of data.
Type	Typically an integer or numeric type (int, float, etc.).	A special marker or "pointer to nowhere" used across various data types.
Arithmetic	Works normally in calculations (e.g., $5 + 0 = 5$).	Generally results in NULL (e.g., $5 + \text{NULL} = \text{NULL}$) because the result of an operation on an unknown is itself unknown.
Logic	Evaluates to false in many languages but is a valid, comparable value.	Often requires special operators (like IS NULL in SQL) because it cannot be compared with standard operators.

3. How do you check NULL values?

Ans: Use the specialized logical operators designed for NULL states.

`SELECT * FROM TableName WHERE ColumnName IS NULL;`

4. Can primary key contain NULL?

Ans: A primary key cannot contain NULL values.

5. Why is NULL important?

Ans: In SQL, NULL is a special marker used to represent missing, unknown, or inapplicable data. It is fundamentally different from a zero (0) or an empty string (""), which are considered actual values.

Topic: **Primary Key**

Technology: SQL

1. What is a primary key?

Ans: In SQL, a primary key is a constraint that uniquely identifies each record (row) in a database table.

2. Why must a primary key be unique?

Ans: A primary key must be unique because its fundamental purpose is to serve as the sole, unambiguous identifier for every record in a database table.

3. Can a table exist without a primary key?

Ans: Yes, a table can exist without a primary key in SQL.

4. What happens if duplicate primary key is inserted?

Ans: Inserting a duplicate primary key results in a constraint violation error, which stops the database from adding the new record to ensure data integrity.

5. Difference between primary key and unique key?

Ans:

Feature	Primary Key	Unique Key
Allows NULLs	No (NOT NULL)	Yes (usually 1 or more)
Allowed per table	Only 1	Multiple
Index type	Clustered	Non-Clustered
Purpose	Entity Integrity (Unique Identity)	Unique constraints

Topic: **Relational Database Concept**

Technology: SQL

1. What is a relational database?

Ans: A type of database that stores and organizes data into tables (also called "relations") consisting of rows and columns.

2. What is the difference between primary key and foreign key?

Ans:

PRIMARY KEY	FOREIGN KEY
A primary key is used to ensure data in the specific column is unique.	A foreign key is a column or group of columns in a relational database table that provides a link between data in two tables.
It uniquely identifies a record in the relational database table.	It refers to the field in a table which is the primary key of another table.
Only one primary key is allowed in a table.	Whereas more than one foreign key is allowed in a table.

It is a combination of UNIQUE and Not Null constraints.	It can contain duplicate values and a table in a relational database.
It does not allow <u>NULL values</u> .	It can also contain NULL values.
Its value cannot be deleted from the parent table.	Its value can be deleted from the child table.
It constraint can be implicitly defined on the temporary tables.	It constraint cannot be defined on the local or global temporary tables.

3. Why do we use relationships between tables?

Ans: Relationships between tables are used to create a structured, efficient, and reliable database.

4. What happens if there is no primary key?

Ans: The table may allow duplicate rows.

5. Explain normalization in relational databases.

Ans: Normalization in SQL is the systematic process of organizing a relational database to reduce data redundancy (duplicate information) and improve data integrity (accuracy and consistency).

Topic: **Selecting Data**

Technology: SQL

1. What is SELECT command?

Ans: The SELECT command is a fundamental SQL statement used to retrieve and display data from one or more tables in a database.

2. What is the use of WHERE clause?

Ans: The WHERE clause in SQL is used to filter records. It ensures that a query only processes or returns rows that meet a specific condition.

3. Difference between SELECT * and SELECT column name?

Ans: In SQL, the difference between SELECT * and SELECT [column_names] centers on efficiency, performance, and code stability. While SELECT * is a convenient shorthand for retrieving all columns, explicitly naming columns is the

standard best practice for production environments.

4. How do you sort data in SQL?

Ans: In SQL, data is sorted using the ORDER BY clause at the end of a SELECT statement.

5. Can we select data from multiple tables?

Ans: Yes, we can select data from multiple tables in SQL.

Topic:**Sorting Data using ORDER BY**

Technology: SQL

1. What is ORDER BY in SQL?

Ans: In SQL, the ORDER BY clause is used to sort the result set of a query in either ascending or descending order.

2. Difference between ASC and DESC?

Ans:

Feature	ASC (Ascending)	DESC (Descending)
Sort Order	Smallest to largest.	Largest to smallest.
Alphabetical	A to Z.	Z to A.
Dates	Oldest to newest.	Newest to oldest.
Default	Yes. If no keyword is specified, SQL defaults to ASC.	No. Must be explicitly stated.

3. What is default sorting order?

Ans: Numeric Values: Rows are sorted from the lowest value to the highest value.

- Text/String Values: Rows are sorted alphabetically (A to Z).
- Changing the Order: To reverse the default and sort in descending order, you must explicitly use the [DESC keyword](#).
- Multiple Columns: When sorting by multiple columns, each column will default to ascending unless otherwise specified. For example, ORDER BY Country, CustomerName sorts both columns in ascending order

4. Can we sort using multiple columns?

Ans: Yes, we can sort by multiple columns in SQL using the ORDER BY clause.

5. Does ORDER BY work with text data?

Ans: Yes, ORDER BY works with text data in SQL. When used on a column containing strings, it sorts the results alphabetically by default.

Topic:**Subqueries Basics**

Technology: SQL

1. What is a subquery?

Ans: A subquery in SQL is a query nested inside another SQL statement, such as SELECT, INSERT, UPDATE, or DELETE. It is also known as an inner query or nested query, while the statement containing it is called the outer query.

2. Where can subqueries be used?

Ans: Subqueries (also known as nested or inner queries) can be used in almost every part of a standard SQL statement where an expression is allowed. They are primarily used to filter, compute, or define data for a larger outer query.

3. Difference between single-row and multi-row subquery?

Ans:

Feature	Single-Row Subquery	Multi-Row Subquery
Return Count	Returns exactly one row (and typically one column).	Returns more than one row.
Operators Used	Comparison operators: =, >, <, >=, <=, <>.	Set comparison operators: IN, ANY, ALL, EXISTS.
Error Handling	Throws an error (e.g., ORA-01427) if it unexpectedly returns multiple rows.	Designed to handle multiple values; returns results without error if multiple rows match.
Typical Use Case	Finding a specific value like MAX(Salary) or AVG(Price).	Checking if a value exists within a list or set of results.

4. What is nested subquery?

Ans: In SQL, a nested subquery (also known as a subquery or inner query) is a query written inside another SQL statement.

5. Can subquery be used in SELECT clause?

Ans: Yes, a subquery can be used in the SELECT clause of a SQL statement.

Topic: **Tables, Rows, and Columns**

Technology: SQL

1. What is the difference between row and column?

Ans: Row: Also known as a record or tuple, a row represents a single, complete [structured data item](#) in a table. For example, in a Customers table, one row contains all information (ID, Name, Email) for one specific customer.

- Column: Also known as a field or attribute, a column is a vertical entity that holds a specific [category of data](#) for every record in the table. For example, the "Email" column contains only email addresses.
- Orientation:
 - Row: Arranged horizontally from [left to right](#).

- Column: Arranged vertically from [top to bottom](#).

2. What is a table in a database?

Ans: A database, a table is a fundamental object used to store data in a structured, two-dimensional format consisting of rows and columns.

3. Can a table exist without columns?

Ans: A table cannot exist without columns in most SQL databases

4. What is a record?

Ans: A record (also commonly called a row) is a single, horizontal entry in a database table that represents a unique item or entity.

5. How is a database different from a table?

Ans: A database is a container that stores and organizes all related data, while a table is a specific structure within that database used to hold actual data records.

Topic:**Types of Databases**

Technology: SQL

1. What is the difference between SQL and NoSQL databases?

Ans:

Aspect	SQL (Relational)	NoSQL (Non-relational)
Data Structure	Tables with rows and columns	Document-based, key-value, column-family, or graph-based
Schema	Fixed schema (predefined structure)	Flexible schema (dynamic and adaptable)
Scalability	Vertically scalable (upgrading hardware)	Horizontally scalable (adding more servers)

Data Integrity	ACID-compliant (strong consistency)	BASE-compliant (more available, less consistent)
Query Language	SQL (Structured Query Language)	Varies (e.g., MongoDB uses its own query language)
Performance	Efficient for complex queries and transactions	Better for large-scale data and fast read/write operations
Use Case	Best for transactional systems (banking, ERP, etc.)	Ideal for big data, real-time web apps, and data lakes
Examples	MySQL, PostgreSQL, Oracle, MS SQL Server	MongoDB, Cassandra, CouchDB, Neo4j

2. Where do we use relational databases?

Ans: They are used in environments where data accuracy, consistency, and the ability to link complex information are critical.

Example : Banking and Finance, E-commerce, Enterprise Systems (ERP/CRM), Healthcare, and Inventory Management.

3. What is a hierarchical database?

Ans: A "hierarchical database" typically refers to the representation and querying of tree-like data structures within a standard relational database.

4. Why is NoSQL used for social media apps?

Ans: NoSQL is used for social media apps because it handles massive volumes of unstructured data, provides high-speed read/write capabilities (low latency), and offers flexible scaling to accommodate rapid, viral user growth.

5. What is the difference between hierarchical and network database?

Ans:

Hierarchical Data Model	Network Data Model
In this model, to store data hierarchy method is used.	In this model, you could create a network that shows how data is related to each other.
It implements 1:1 and 1:n relations.	It implements 1:1, 1:n and also many to many relations.
To organize records, it uses tree structure.	To organize records, it uses graphs.
Records are linked with the help of pointers.	Records are linked with the help of linked list.
Insertion anomaly exists in this model i.e. child node cannot be inserted without the parent node.	There is no insertion anomaly.
Deletion anomaly exists in this model i.e. it is difficult to delete the parent node.	There is no deletion anomaly.
It is used to access the data which is complex and asymmetric.	It is used to access the data which is complex and symmetric.

Topic: **Updating Data**

Technology: SQL

1. What is UPDATE command?

Ans: The SQL UPDATE command is a Data Manipulation Language (DML) statement used to modify existing data within a database table.

2. Why is WHERE clause important in UPDATE?

Ans: The WHERE clause is crucial in an UPDATE statement because it defines which specific rows in a table will be modified.

3. Can we update multiple columns at once?

Ans: Yes, we can update multiple columns in a single SQL UPDATE statement by separating the column-value pairs with commas in the SET clause. This approach allows modifying multiple fields in a single row based on a WHERE condition, or all rows if the condition is omitted.

4. What happens if data type is wrong while updating?

Ans: Updating data with the wrong data type generally results in a Data Type Mismatch error, a runtime error, or unintentional data loss or corruption, depending on whether it occurs in a database, an application, or a spreadsheet.

5. How do you update all rows?

Ans: To update all rows in a SQL table, we can use the UPDATE statement without a WHERE clause. Omitting the WHERE clause tells the database management system to apply the changes to every record in the table.

UPDATE table_name

SET column1 = value1, column2 = value2;

Topic: **What is Database**

Technology: SQL

1. What is a database?

Ans: A database is a structured, electronic collection of organized data stored in a [Relational Database Management System \(RDBMS\)](#). It acts as a digital container for related [information, such as tables, views, and indexes](#), allowing for efficient storage, retrieval, and management of data.

2. What is the difference between row and column?

Ans:

Feature	Row	Column
Common Alias	Record or Tuple	Field or Attribute
Orientation	Horizontal (Left to Right)	Vertical (Top to Bottom)
Represents	A single object or entity	A specific property of all entities

Quantity	Dynamic; grows with new entries	Fixed; defined by the table schema
Data Consistency	Contains multiple data types (e.g., Name, Age, Date)	Contains a single data type for all entries
Example	One specific customer's profile	The "Email Address" field for all customers

3. Where are databases used in real life?

Ans: Database Management Systems (DBMS) are essential tools in daily life, powering everything from banking transactions and e-commerce websites (e.g., Amazon) to social media platforms (e.g., Facebook) and streaming services (e.g., Netflix).

4. What is a relational database?

Ans: A relational database is a type of database that organizes structured data into tables (also called "relations") consisting of rows and columns.

5. How is a database different from Excel?

Ans: A database is designed for secure, multi-user, and high-volume data storage, ensuring high data integrity through structured tables, while Excel is a flexible, cell-based spreadsheet tool optimized for individual data analysis, calculations, and visualization. Databases are better for massive, permanent datasets, whereas Excel is best for smaller, dynamic, and frequently updated projects.

Topic: **WHERE Clause**

Technology: SQL

1. What is WHERE clause in SQL?

Ans: The WHERE clause in SQL is used to filter records and retrieve only those that meet a specific condition.

2. What is the difference between AND and OR?

Ans:

Feature	AND Operator	OR Operator
Logic	Returns TRUE only if all conditions are true.	Returns TRUE if at least one condition is true.
Data Scope	Narrows results by adding more strict filters.	Broadens results by including more possibilities.
Precedence	Higher precedence (evaluated first).	Lower precedence (evaluated after AND).

3. Can we use WHERE without SELECT?

Ans: Yes, we can use the WHERE clause without a SELECT statement. While it is most commonly seen in SELECT queries to filter results, it is also a fundamental part of other Data Manipulation Language (DML) statements.

4. What happens if WHERE is not used?

Ans: SELECT Statement: The query will return all rows from the table rather than a filtered subset.

- UPDATE Statement: Every single record in the table will be modified to the new value(s) provided.
- DELETE Statement: Every record will be removed from the table, effectively emptying it.
- Performance: For large datasets, omitting WHERE can cause performance issues because the database must process the entire table.
- Grouped Data: You can still filter aggregated data using the HAVING clause, which functions similarly to WHERE but is applied after grouping.

5. Which operators are used in WHERE?

Ans: These are used to compare a column value with another value or expression, returning TRUE, FALSE, or UNKNOWN.

- = : Equal to (e.g., WHERE age = 24)
- <> or != : Not equal to (e.g., WHERE amount <> 400)
- > : Greater than (e.g., WHERE age > 21)
- < : Less than (e.g., WHERE amount < 400)
- >= : Greater than or equal to (e.g., WHERE age >= 22)
- <= : Less than or equal to (e.g., WHERE amount <= 400)

Regards
Ashka Singh

----- Forwarded message -----

From: **ashka singh** <sgsinghashka@gmail.com>

Date: Wed, 1 Apr 2026 at 22:51

Subject: Fwd: TA || Ashka || TM- B141

To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topic Data Science Practice, kindly find the requested details below.

Date: 14th March 2026

Topic: Data Science Practice

1. Explain the difference between structured and unstructured data with examples.

Ans: Structure: Structured data follows a strict schema; unstructured data does not.

- Storage: Structured uses relational databases; unstructured uses data lakes or file systems.

- Analysis: Structured is easily analyzed using SQL or BI tools; unstructured requires AI/Machine Learning (like NLP) to extract meaning.
- Volume: Unstructured data makes up about 80% of enterprise data.

Examples:

- Structured Data: A customer CRM database table containing Name, Phone Number, and Address.
- Unstructured Data: A recording of a customer service phone call, a social media post, or a photograph.

2. What are the steps involved in the data science life cycle?

Ans: The data science life cycle is a structured, iterative process used to deliver data science projects effectively. It typically involves seven key stages: understanding the business problem, gathering and cleaning data, conducting exploratory analysis, modeling, evaluating the results, deploying the model, and communicating insights to stakeholders.

3. How do you handle missing data in a dataset? Explain different techniques.

Ans: Handling missing data involves either deleting the incomplete entries or using imputation to fill them with estimated values.

Missing data methods involve identifying patterns—Missing Completely at Random (MCAR), At Random (MAR), or Not at Random (MNAR)—and addressing them through deletion, simple imputation (mean/median), or advanced techniques (multiple imputation, model-based methods). Proper handling avoids biased results, with Multiple Imputation by Chained Equations (MICE) often preferred.

4. What is the difference between pandas Series and DataFrame?

Ans:

Feature	Pandas Series	Pandas DataFrame
Dimensions	1D (one-dimensional)	2D (two-dimensional)
Structure	A single column or row with an index	A table with rows, columns, and both row/column labels
Data Types	Homogeneous (elements must be the same type)	Heterogeneous (columns can have different data types)
Size Mutability	Size-immutable (once created, size cannot be changed easily)	Size-mutable (elements/columns can be added or dropped)
Indexing	Row labels only	Both row and column labels
Functionality	Fewer features, focused on single-column operations	More extensive features for multi-column operations (merging, filtering, etc.)

5. What is Exploratory Data Analysis (EDA) and why is it important?

Ans: Exploratory Data Analysis (EDA) is an approach to analyzing datasets using visualization and summary statistics to understand their main characteristics, identify patterns, detect outliers, and check assumptions.

Reason of Important

1. Reduces Errors: Identifies data issues early, before complex modeling.
2. Validates Assumptions: Confirms whether data meet the assumptions required for specific analytical methods.
3. Informs Decision Making: Provides foundational insights that make subsequent analyses more accurate and actionable
4. Generates Hypotheses: Assists in forming testable hypotheses about the data.

Regards
Ashka Singh

----- Forwarded message -----

From: **ashka singh** <sgsinghashka@gmail.com>
Date: Wed, 1 Apr 2026 at 21:48
Subject: Fwd: TA || Ashka || TM- B141
To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topics of Data Cleaning Basics, Data Filtering, Data Type Conversion, Grouping and Aggregation, Handling Missing Values, Merging and Joining Data, String Data Processing, Correlation Analysis, Data Visualization Principles, Date and Time Handling, Descriptive Statistics, Exploratory Data Analysis (EDA) Workflow, Matplotlib Basics, Reshaping Data (pivot, melt), and Seaborn Basics. Kindly find the requested details below.

Please be apprised that due to personal urgency, I was unable to share the assignment at the scheduled time.

Date: 14th March 2026
Topic: Data Cleaning Basics
Technology: Data Science

1. What is data cleaning?

Ans: Data cleaning (or data cleansing) is the process of detecting, correcting, or removing inaccurate, incomplete, improperly formatted, or duplicated data within a dataset. It transforms "dirty" data into reliable, high-quality information, ensuring consistency and accuracy for analysis, reporting, and machine learning models.

2. Why are missing values a problem?

Ans: Missing values are problematic because they reduce statistical power, introduce bias, and prevent machine learning models from functioning correctly.

3. How do you remove duplicate records?

Ans: Using built-in functions for simple lists, or with the Pandas library for structured data like DataFrames.

4. What are outliers?

Ans: Outliers are data points that differ significantly from the rest of the dataset and do not follow the general pattern. They can occur due to errors, rare events or natural variability in data.

- Appear as unusually high or low values
- Can affect mean, variance, and model performance
- Detected using statistical and visual methods
- Important to analyze before removing or treating them

5. Why is data type correction important?

Ans: Data type correction is critical to ensure computational accuracy, prevent application crashes, and optimize storage/processing performance.

Date: 14th March 2026

Topic: Data Filtering

Technology: Data Science

1. What is data filtering?

Ans: Data filtering is the process of selecting a smaller, relevant subset of data from a larger dataset based on specific rules, criteria, or conditions to make analysis faster, cleaner, and more accurate.

2. How do you apply multiple conditions in Pandas?

Ans: Use parentheses around each condition and combine them with bitwise operators: & (and), | (or), and ~ (not).

```
# Filter rows where 'Age' > 25 AND 'Status' is 'Active'
filtered_df = df[(df['Age'] > 25) & (df['Status'] == 'Active')]
```

```
# Filter rows where 'Age' > 25 OR 'Status' is 'Active'
filtered_df = df[(df['Age'] > 25) | (df['Status'] == 'Active')]
```

3. What does isin() function do?

Ans: The isin() function checks if elements in a data structure (like a list, Series, or DataFrame) exist within a specified collection of values. It returns a Boolean (True/False) result.

4. How to filter rows based on a text column?

Ans: Using a simple AutoFilter, a specific formula, or in programming languages like Python (pandas).

5. Why is filtering important before analysis?

Ans: Filtering is essential because it removes noise, errors, and irrelevant information, ensuring analysis is based on accurate, reliable, and high-value data.

Date: 14th March 2026

Topic: Data Type Conversion

Technology: Data Science

1. What is data type conversion?

Ans: Data type conversion (or typecasting) is the process of changing data from one type (e.g., string, integer, float) to another.

2. Why is data type conversion required?

Ans: Data type conversion is required primarily to ensure compatibility between different data formats so they can interact correctly.

3. How do you convert column type in Pandas?

Ans: We can convert column types in Pandas primarily using the `.astype()` method, or with specialized functions like `pd.to_numeric()`, `pd.to_datetime()`, or `df.convert_dtypes()` for specific use cases.

4. How to convert a string to a date?

Ans: To convert a string to a date, we must use language-specific functions that parse the string based on its exact format.

from datetime import datetime

```
date_string = '2023-10-30 12:00:00'
format_string = '%Y-%m-%d %H:%M:%S'
date_object = datetime.strptime(date_string, format_string)
```

Output: 2023-10-30 12:00:00

5. What problems occur if data type is wrong?

Ans: Using the wrong data type (e.g., treating numbers as text, or dates as strings) creates significant data quality issues, ranging from minor functional bugs to major operational failures and financial losses.

Date: 14th March 2026

Topic: Grouping and Aggregation

Technology: Data Science

1. What is grouping in data analysis?

Ans: The process of organizing raw, individual data points into smaller, meaningful categories or intervals.

2. What does `groupby()` do?

Ans: The `pandas.DataFrame.groupby()` method is a core function for data analysis that implements the "split-apply-combine" strategy. It allows us to organize data into groups based on one or more criteria (e.g., specific column values) and then perform operations on each group independently.

3. Difference between `sum()` and `mean()`?

Ans: `sum()` calculates the total of all values in a dataset, while `mean()` (average) calculates the arithmetic center by taking that total and dividing it by the number of values.

4. How to group by multiple columns?

Ans: To group a `pandas.DataFrame` by multiple columns, we pass a list of the column names to the `groupby()` method. After grouping, we typically apply an aggregation function like `sum()`, `mean()`, `count()`, or `agg()` to summarize the data within each group.

5. What is `agg()` used for?

Ans: The `agg()` method in Python's Pandas library is used to apply one or more aggregation functions (like `sum`, `mean`, `max`, `min`) to a `DataFrame` or `Series`.

Date: 14th March 2026
Topic: Handling Missing Values
Technology: Data Science

1. What are missing values?

Ans: A missing value, or missing data, occurs when no data is recorded for a variable in an observation within a dataset. Common synonyms include **NaN (Not a Number)**, **Null**, **None**, or blank cells.

2. How do you detect missing values?

Ans: Use `df.isnull().sum()` in Python's Pandas library to get a count of missing values per column. For a visual overview, use `missingno.matrix(df)` or a Seaborn heatmap to identify patterns in missing data.

3. When should you remove missing data?

Ans: Remove missing data only when the missingness is random (MCAR), the amount is negligible (typically <5%), or the remaining dataset is large enough that removing rows/columns will not introduce bias.

4. What is the difference between `dropna()` and `fillna()`?

Ans: `dropna()` removes rows or columns containing missing values (NaN), while `fillna()` replaces missing values with specified data (like 0, a mean value, or a string). Use `dropna()` when missing data makes a row/column useless; use `fillna()` when you want to preserve your dataset size by imputing missing entries.

5. Why do we use mean or median to fill missing values?

Ans: Use mean or median to fill missing values because they provide statistically representative estimates that preserve the dataset's central tendency without introducing the extreme biases caused by arbitrary placeholders.

Date: 14th March 2026
Topic: Merging and Joining Data
Technology: Data Science

1. What is data merging?

Ans: Data merging is the process of combining two or more datasets into a single, unified dataset, usually based on a common key, column, or identifier.

2. Difference between inner and outer join?

Ans: An inner join combines two DataFrames and returns only the rows with matching values in the specified key column(s), similar to a set intersection. An outer join, conversely, returns all rows from both DataFrames, combining matching rows and filling non-matching fields with NaN (Not a Number) values, acting like a set union.

3. What is left join?

Ans: A left join in Python, typically performed using the [pandas library's merge\(\)](#) or [.join\(\)](#) functions, combines two DataFrames by keeping all rows from the "left" DataFrame and only the rows from the "right" DataFrame that have a matching key.

4. How to merge two datasets in Pandas?

Ans: To merge two datasets (DataFrames) in Pandas, the primary functions are `pd.merge()`, `.join()`, and `pd.concat()`. The best choice depends on how you want to combine the data (e.g., based on common columns, indices, or simply stacking them).

5. What happens if key column has no match?

Ans: `how='inner'` (Default): Rows with no match in the other DataFrame are excluded from the result. Only rows with matching keys in both DataFrames are kept.

- `how='left'`: All rows from the left DataFrame are kept. For rows where no match is found in the right DataFrame, the columns from the right DataFrame will contain NaN (Not a Number) values.
- `how='right'`: All rows from the right DataFrame are kept. For rows where no match is found in the left DataFrame, the columns from the left DataFrame will contain NaN values.

`how='outer'`: All rows from both DataFrames are kept. Unmatched columns for any given row are filled with NaN values.

Date: 14th March 2026

Topic: Removing Duplicates

Technology: Data Science

1. What are duplicate records?

Ans: Duplicate records refer to identical data entries that appear more than once within a dataset, such as multiple identical rows in a pandas DataFrame or repeated items in a list.

2. How do you detect duplicates in Pandas?

Ans: Use `df.duplicated()` to identify duplicate rows. It returns a Boolean Series where True marks a duplicate. Filter rows with `df[df.duplicated()]` to view them. Use the `subset` parameter to check specific columns and keep control of which instances (first, last, or all) are marked as duplicates.

3. How do you remove duplicates from a specific column?

Ans: The most efficient method is using the `drop_duplicates()` function from the [pandas library](#) with the `subset` parameter.

4. What does `keep="last"` do?

Ans: In Pandas, `keep='last'` is a parameter used within the `drop_duplicates()` method to determine which duplicate row to retain. Specifically, `keep='last'` tells Pandas to keep the final occurrence of a set of duplicate values and remove all previous occurrences.

5. Why is removing duplicates important?

Ans: Ensuring data accuracy and improving performance.

Date: 14th March 2026

Topic: Sorting Data

Technology: Data Science

1. What is data sorting?

Ans: The process of arranging elements within a collection (like a list, tuple, or dictionary) into a specific sequence, such as numerical or alphabetical order.

2. How do you sort data in descending order?

Ans: In Python, we can sort data in descending order using the built-in `sort()` method (for lists) or the `sorted()` function (for any iterable), by setting the `reverse` parameter to `True`.

3. How to sort using multiple columns?

Ans: The `sort_values()` method is used for multi-column sorting in pandas.

Multiple Columns/Orders: Pass a list of columns to `by` and a corresponding list of booleans to `ascending` (e.g., `ascending=[True, False]`) to mix ascending and descending sorts.

4. What is default sorting order?

Ans: The default sorting order for both the `sort()` method (for lists) and the `sorted()` function (for any iterable) is ascending order. This applies to various data types.

- Numbers are sorted from smallest to largest (e.g., [1, 5, 9] from [5, 1, 9]).
- Strings are sorted alphabetically (lexicographical order). By default, uppercase letters come before lowercase letters (based on ASCII values), but this can be adjusted with a key function.

5. Difference between `sort_values()` and `sort_index()`?

Ans: In pandas, `sort_values()` sorts a DataFrame or Series based on the content of its columns, while `sort_index()` sorts the data based on its index labels (row or column labels).

Date: 14th March 2026

Topic: String Data Processing

Technology: Data Science

1. What is string data processing?

Ans: String data processing in Python is the act of creating, modifying, analyzing, and transforming textual data using Python's built-in `str` class methods, operators, and functions. Strings are sequences of immutable Unicode characters, and operations always return new strings instead of modifying the original in place.

2. Why do we use `strip()` function?

Ans: The Python `strip()` function is used to remove unwanted leading and trailing characters from a string.

3. How to convert column text to lowercase in Pandas?

Ans: To convert text within a specific column to lowercase in pandas, the most common and efficient method is using the `.str.lower()` accessor and assigning the result back to the same column.

4. How do you check if email contains certain text?

Ans: Use the `in` operator for simple checks or the `re` module (regular expressions) for more complex pattern matching.

5. Why is string cleaning important in data analysis?

Ans: It directly ensures the accuracy, consistency, and reliability of the data, which in turn leads to valid insights and sound decision-making.

Date: 15th March 2026

Topic: Correlation Analysis

Technology: Data Science

1. What is correlation analysis?

Ans: Primarily performed using the [Pandas](#), [NumPy](#), and [SciPy](#) libraries.

2. Difference between positive and negative correlation?

Ans: Positive Correlation: Variables move in the same direction, where an increase in one generally corresponds to an increase in the other, resulting in a coefficient.

- between 0 and +1. A perfect relationship is +1. Example: Height and weight.
- Negative Correlation: Variables move in opposite directions, where an increase in one corresponds to a decrease in the other, yielding a coefficient
-  6075f0e5-5ad5-44c8-a34c-32467cca6ae4.jpg
- between 0 and -1. A perfect relationship is -1.

3. What does correlation value 0 mean?

Ans: A correlation value of **0** (or very close to 0) in Python—typically calculated using `pandas.DataFrame.corr()`, `numpy.corrcoef()`, or `scipy.stats.pearsonr()`—means there is no linear relationship between the two variables being compared

4. How do you calculate correlation in Pandas?

Ans: Correlation in Pandas is primarily calculated using the `.corr()` method on a `DataFrame` or `Series`. This function calculates the pairwise correlation coefficients between numerical columns, excluding missing values (NA/null).

5. Why does correlation not imply causation?

Ans: Python libraries can only calculate a statistical relationship (correlation); they cannot inherently understand the real-world context, directionality, or the potential for other variables that might explain the observed link.

Date: 15 March 26

Topic: Data Visualization Principles

Technology: Data Science

1. What are data visualization principles?

Ans: It's effectively and ethically communicating insights from data using libraries like Matplotlib, Seaborn, and Plotly.

2. Why is chart selection important?

Ans: Choosing the right chart in Python (and data visualization generally) is crucial for effective communication of insights, enabling faster and more informed decision-making, and ensuring the data is presented clearly and accurately.

3. What mistakes should be avoided in charts?

Ans: Manipulating axes (especially not starting at zero), using inappropriate chart types, information overload (chartjunk), poor color choices, and missing context/labels.

4. When should scatter plot be used?

Ans: Use a scatter plot when you need to analyze the relationship between two continuous numeric variables.

5. Why is clarity important in visualization?

Ans: It bridges the gap between raw data and actionable insight. Without clarity, information is easily misinterpreted, ignored, or dismissed.

Date: 15 March 26

Topic: Date and Time Handling

Technology: Data Science

1. What is date-time handling?

Ans: Dates and times are primarily handled using the built-in `datetime` module, which provides classes for representing and manipulating temporal data.

2. How do you convert a column to datetime?

Ans: To convert a column to datetime, the most effective method is using the `pd.to_datetime()` function from the [Pandas library](#). This function can automatically parse many common date formats.

3. How to extract year from date column?

Ans: The pandas library with the `.dt.year` accessor. This requires ensuring the column is first converted to a `datetime` object using `pd.to_datetime()`.

4. How to calculate difference between two dates?

Ans: Calculate the difference between two dates by subtracting one `datetime` or `date` object from another. The result is a `timedelta` object, which represents the duration between the two dates.

5. Why is date-time important in data analysis?

Ans: Date-time data is critical because it provides context, sequence, and duration to raw information.

Date: 15 March 26

Topic: Descriptive Statistics

Technology: Data Science

1. What is descriptive statistics?

Ans: Descriptive statistics generally fall into two main categories, both readily calculable using Python's libraries such as [pandas](#), [NumPy](#), and the built-in [statistics](#) module.

2. Difference between mean and median?

Ans: The mean is the arithmetic average of a dataset, calculated by summing all values and dividing by the count of values. The median is the middle value of an ordered dataset, which is less affected by extreme values (outliers).

3. What is standard deviation?

Ans: Standard deviation is a statistical measure of data dispersion from the mean, typically calculated using functions from the `statistics`, `numpy`, or `pandas` libraries. A low standard deviation means values are clustered around the mean, while a high standard deviation indicates values are spread out over a wider range.

4. When should median be preferred over mean?

Ans: When the dataset is skewed or contains outliers (extreme values).

5. What does `describe()` function show?

Ans: The `describe()` function in the Python **pandas** library generates descriptive statistics that summarize the central tendency, dispersion, and shape of a dataset's distribution. By default, when applied to a `pandas DataFrame`, it provides a statistical summary for only the **numeric columns**, but it can also be customized to analyze categorical data or all data types.

Date: 15 March 26

Topic: Exploratory Data Analysis (EDA) Workflow

Technology: Data Science

1. What is EDA?

Ans: Exploratory Data Analysis (EDA) is the process of using libraries like [Pandas](#), [NumPy](#), [Matplotlib](#), and [Seaborn](#) to analyze and visualize datasets, summarize their main characteristics, and uncover patterns, anomalies, and relationships between variables.

2. Why is EDA important?

Ans: It serves as the foundational "detective work" for any data project, ensuring you thoroughly understand the data's characteristics, spot potential problems, and make informed decisions before building models.

3. What are steps in EDA?

Ans: The core process involves importing libraries and data, performing initial inspection and cleaning, carrying out univariate, bivariate, and multivariate analysis using descriptive statistics and visualizations, and finally, documenting the

findings.

4. Which tools are used in EDA?

Ans: Use Python (libraries: Pandas, NumPy, Matplotlib, Seaborn, Plotly) or R for data analysis and visualization. For no-code or business intelligence workflows, use Power BI, Tableau, or KNIME.

5. How does EDA improve model accuracy?

Ans: EDA improves model accuracy by enabling informed data cleaning, strategic feature engineering, and appropriate algorithm selection.

Date: 15 March 26

Topic: Matplotlib Basics

Technology: Data Science

1. What is Matplotlib?

Ans: Matplotlib is a comprehensive, open-source library for the Python programming language used to create static, animated, and interactive visualizations.

2. How do you create a line chart?

Ans: To create a line chart in [Matplotlib](#), you use the `pyplot.plot()` function to connect data points with a line. The basic process involves importing the library, defining your data points for the x and y axes, plotting the data, and then displaying the chart.

3. Why is `plt.show()` important?

Ans: `plt.show()` is important because it displays all created figures and manages the interactive process required for the plot windows to appear and stay open.

4. Difference between bar chart and line chart?

Ans: Line graphs use connected points, whereas bar graphs use rectangular bars to show magnitude.

5. How to add labels to a chart?

Ans: The most common way to add labels to the x and y axes, and a title to the entire plot, is by using the `xlabel()`, `ylabel()`, and `title()` functions from the pyplot interface.

Date: 15 March 26

Topic: Reshaping Data (pivot, melt)

Technology: Data Science

1. What is data reshaping?

Ans: Data reshaping is the process of reorganizing a dataset's structure—rows and columns—without changing the data itself.

2. Difference between `pivot()` and `melt()`?

Ans: `pivot()` transforms data from long to wide format (increasing columns, decreasing rows). `melt()` transforms data from wide to long format (decreasing columns, increasing rows).

3. When should pivot be used?

Ans: Use pivot when your goals or data analysis require a fundamental shift in direction or perspective. Use the `.pivot()` method when you need simple data reshaping and your data has unique index/column combinations. It does not support aggregation and will raise an error if it encounters duplicate entries for the specified row and column pair.

4. Why do we convert wide data to long format?

Ans: We convert wide data to long format to make it "tidy" and machine-readable.

5. What parameters does pivot() need?

Ans : In pandas (df.pivot()), the function requires:

- index: The column(s) to use as the row labels in the new DataFrame.
- columns: The column(s) whose unique values will become the new column headers.
- values (*Optional*): The column(s) to populate the cells of the new DataFrame. If omitted, all remaining columns are used, resulting in hierarchical columns.

Date: 15 March 26

Topic: Seaborn Basics

Technology: Data Science

1. What is Seaborn?

Ans: Seaborn is a Python library for creating attractive, informative statistical graphics.

2. Difference between Matplotlib and Seaborn?

Ans: [Matplotlib](#) is a low-level, general-purpose library that offers fine-grained control over all plot elements, while Seaborn is a higher-level library built on top of Matplotlib.

3. When to use box plot?

Ans: Use box plots (box-and-whisker plots) to visualize the distribution, spread, and skewness of numerical data, particularly when comparing multiple groups or identifying outliers.

4. How do you create histogram in Seaborn?

Ans: To create a histogram in Seaborn, we primarily use the `sns.histplot()` function, which is the recommended function in newer versions of the library. This function is built on top of Matplotlib and allows for easy creation of informative statistical graphics.

5. Why is Seaborn preferred for statistical plots?

Ans: It automates complex statistical calculations (like confidence intervals and regressions), provides a concise, dataset-oriented API that integrates natively with pandas, and produces professional-grade aesthetics by default.

Regards

Ashka Singh

----- Forwarded message -----

From: **ashka singh** <sgsinghashka@gmail.com>

Date: Sat, 14 Mar 2026 at 18:41

Subject: Fwd: TA || Ashka || TM- B141

To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topics of Data Collection Techniques, Data Loading (CSV, Excel, JSON, Data Science Life Cycle, Data Science Tools Overview, Introduction to Data Science, Data Sources (Databases, APIs, Files), NumPy Arrays, Python Libraries for Data Analysis, Array Operations, Data Inspection (head, info, describe), Data Loading (CSV, Excel, JSON) , Pandas DataFrame, P and andas Series, kindly find the requested details below.

Date: 7th March 2026

Topic: **Data Collection Techniques**

Technology: Data Science

1. What is data collection?

Ans: Data collection is the systematic process of gathering, measuring, and analyzing accurate information from various sources to answer research questions, solve problems, or make data-driven decisions.

2. Name some data collection techniques.

Ans: Data collection techniques include surveys/questionnaires, interviews (structured or unstructured), observations, focus groups, and document analysis.

3. What is web scraping?

Ans: Web scraping is the automated process of extracting large amounts of structured or unstructured data from websites using software tools.

4. Why is data accuracy important?

Ans: Data accuracy is critical because it ensures that information, analytics, and AI models reflect real-world conditions, allowing for reliable, fact-based decision-making.

5. What ethical issues can arise in data collection?

Ans: Ethical issues in data collection revolve around respecting individual rights, ensuring transparency, and maintaining fairness. Key issues include data privacy violations, lack of informed consent, security breaches, bias, and the potential for harm or misuse.

Date: 7th March 2026

Topic: **Data Loading (CSV, Excel, JSON)**

Technology: Data Science

1. How do you load CSV file in Python?

Ans: We can load a CSV file in [Python](#) using either the built-in `csv` module (ideal for basic reading/writing without external dependencies) or the `pandas` library (recommended for data analysis and manipulation due to its powerful `DataFrame` structure).

2. Difference between CSV and Excel file?

Ans: CSV files are plain text, lightweight, and ideal for raw data storage and data exchange across platforms. [Excel files \(.xlsx/.xls\)](#) are proprietary binary files supporting multiple sheets, formatting, and formulas.

3. Where is JSON mostly used?

Ans: JSON is predominantly used as the universal data interchange format across the internet, primarily in web APIs, configuration files, and databases.

4. Why do we use head() after loading data?

Ans: The .head() method is used after loading data to get a quick overview of its contents and structure without printing the entire dataset.

5. What problems can occur while loading data?

Ans: Data loading issues frequently include schema mismatches (e.g., string into integer), data format errors, missing mandatory fields, and character set incompatibility.

Date: 7th March 2026

Topic: **Data Science Life Cycle**

Technology: Data Science

1. What are the stages of Data Science Life Cycle?



Ans :

2. Why is data cleaning important?



Ans:

3. What happens in deployment stage?



4. How do you evaluate a model?

Ans: Model evaluation assesses a machine learning model's performance, ensuring it generalizes well to new data rather than just memorizing training data.

5. Why is monitoring required after deployment?

Ans: Monitoring is essential after deployment to ensure software or AI models continue to perform reliably, maintain high accuracy, and handle real-world, dynamic data.

Date: 7th March 2026

Topic: **Data Science Tools Overview**

Technology: Data Science

1. What are common tools in Data Science?



2. Why is Python widely used?

Ans: Python is widely used due to a combination of its simple, readable syntax, versatility across many domains, and a vast ecosystem of libraries and strong community support.

3. Difference between Power BI and Python?

Ans: Power BI is a user-friendly business intelligence (BI) tool for visualization and reporting, while Python is a versatile programming language for a wide range of tasks, including complex data analysis, automation, and machine learning.

4. What is TensorFlow used for?

Ans: TensorFlow is an end-to-end, open-source platform for building, training, and deploying machine learning (ML) models, particularly deep learning applications.

5. When do we need Big Data tools?

Ans: Big Data tools are required when data volume, velocity, or variety exceeds the capacity of traditional databases (SQL) and software, typically in the terabyte or petabyte range.

Date: 7th March 2026

Topic: **Data Sources (Databases, APIs, Files)**

Technology: Data Science

1. What are different data sources?



2. Difference between API and Database?

Ans: An API (Application Programming Interface) is a set of rules that allow different software applications to communicate and exchange data, acting as a messenger or intermediary. A database is an organized system for storing, managing, and retrieving structured data.

3. When should we use CSV files?

Ans: CSV files should be used for storing, transferring, or exchanging simple, structured, and tabular data across different systems, applications, or platforms.

4. How do APIs help in real-time systems?

Ans: APIs (Application Programming Interfaces) are crucial for real-time systems by enabling instant, seamless communication and data exchange between different software applications.

5. Which data source is best for large structured data?

Ans: For large, structured data, a [Data Warehouse](#) (e.g., Snowflake, Amazon Redshift, Azure Synapse Analytics).

Date: 7th March 2026

Topic: **Introduction to Data Science**

Technology: Data Science

1. What is Data Science?

Ans: Data science is the study of data used to extract meaningful insights for business decisions. It combines mathematics, computing and domain knowledge to solve real-world problems and uncover hidden patterns.

2. What are the steps in Data Science process?

Ans: Defining the problem, collecting and cleaning data, exploratory data analysis (EDA), modeling, and deployment/communication.

3. Difference between structured and unstructured data?



4. Why is data cleaning important?

Ans: Data cleaning is critical because it ensures data accuracy, consistency, and reliability, which directly impacts decision-making and operational efficiency. By removing duplicates, fixing errors, and standardizing formats, it prevents faulty insights, enhances machine learning model performance, saves time, reduces costs, and ensures regulatory compliance.

5. What tools have you used in Data Science?

Ans: Python (Pandas, NumPy, Scikit-learn), SQL for database management, and visualization software like Tableau or PowerBI.

Date: 7th March 2026

Topic: **NumPy Arrays**

Technology: Data Science

1. What is NumPy?

Ans: A foundational open-source library for Python designed for high-performance scientific computing and data analysis.

2. Difference between list and NumPy array?

Ans: In Python, a list is a built-in, general-purpose data structure, while an array (from the built-in array module or the popular NumPy library) is a specialized, memory-efficient structure optimized for numerical operations. The core difference lies in their data type flexibility and performance characteristics.

3. What does arr.shape return?

Ans: `arr.shape` returns a **tuple of integers** that indicates the number of elements (size) along each dimension of a NumPy array.

For a 1D array (e.g., 5 elements): It returns `(5,)`.

For a 2D array (e.g., 2 rows, 3 columns): It returns `(2, 3)`.

For a 3D array (e.g., 2 blocks, 3 rows, 4 columns): It returns `(2, 3, 4)`.

4. What are advantages of NumPy?

Ans: NumPy offers superior performance for numerical data, operating 10–100x faster than standard Python lists through vectorized C-based operations and contiguous memory allocation.

5. How do you create a 2D array?

Ans: A 2D array, or matrix, is created by nesting arrays within an array to form rows and columns.

Date: 7th March 2026

Topic: Python Libraries for Data Analysis

Technology: Data Science

1. What is Pandas used for?

Ans: Pandas is a fast, powerful, and flexible open-source Python library used for data manipulation, cleaning, and analysis.

2. Difference between NumPy and Pandas?

Ans: NumPy excels at fast, numerical computations on homogeneous multi-dimensional arrays, while Pandas is designed for data analysis and manipulation using labeled, structured tabular data (DataFrames and Series).

3. Which library is used for visualization?

Ans: [Matplotlib](#) is the most popular, fundamental Python library for creating static, interactive, and animated visualizations. Other top choices include Seaborn (statistical graphics), [Plotly](#) (interactive web plots), [Bokeh](#) (large dataset dashboards), and [Vega-Altair](#) (declarative, concise code).

4. What is Scikit-learn?

Ans: [Scikit-learn](#) (also known as sklearn) is a widely used open-source Python library for machine learning.

5. Why are Python libraries important?

Ans: It provide reusable, pre-written code that allows developers to perform complex tasks efficiently without having to write everything from scratch.

Date: 8th March 2026

Topic: Array Operations

Technology: Data Science

1. What are array operations in NumPy?

Ans: NumPy array operations are highly optimized functions and methods for efficiently manipulating and performing mathematical computations on N-dimensional array objects ().

2. Difference between array + number and array + array?

Ans: array + number adds the number to every single element in the array.

- **array + array performs element-by-element addition of corresponding elements in the two arrays (which must typically be the same size).**

3. What does reshape() do?

Ans: [numpy.reshape\(\)](#) changes the shape of a NumPy array (dimensions and size) without altering its data, allowing a 1D array to become a 2D or 3D array, or flattening multidimensional arrays.

4. How is NumPy faster than normal Python loops?

Ans: NumPy is significantly faster than normal Python loops primarily due to vectorization.

5. Name some statistical operations in NumPy.

Ans: NumPy provides a comprehensive suite of functions for performing statistical operations on arrays, which are essential for data analysis and scientific computing.

Key statistical operations available in NumPy include:

Measures of Central Tendency

`numpy.mean()` : Computes the arithmetic mean (average) of the array elements.

`numpy.median()` : Calculates the median (middle value of a sorted array).

`numpy.average()` : Computes the weighted average of elements in an array, useful when different data points have different importance.

Date: 8th March 2026

Topic: **Data Inspection (head, info, describe)**

Technology: Data Science

1. What is data inspection?

Ans: Data inspection is the crucial preliminary step in data analysis where raw data is examined for quality, structure, and accuracy before cleaning or modeling.

2. Difference between head() and describe()?

Ans: The primary difference is that the head() method is used for viewing the first few rows of a DataFrame, while the describe() method is used for generating a statistical summary of the data.

3. What information does info() provide?

Ans: The [pandas info\(\) method](#) provides a concise summary of a DataFrame.

4. Why is data inspection necessary?

Ans: Data inspection is necessary to ensure data quality, accuracy, and reliability before analysis, preventing "garbage in, garbage out" scenarios.

5. How do you detect missing values in dataset?

Ans: Using libraries like Pandas in Python. Missing values are usually represented as NaN (Not a Number), None, or empty cells, and they must be identified to ensure data quality and model performance.

Date: 8th March 2026

Topic: **Data Loading (CSV, Excel, JSON)**

Technology: Data Science

1. How do you load CSV file in Python?

Ans: We can load a CSV file in [Python](#) using either the built-in csv module (standard library) or the popular external pandas library.

2. Difference between CSV and Excel file?



Ans:

3. Where is JSON mostly used?

Ans: JSON is predominantly used in web and mobile application development as the standard format for data exchange and in the implementation of APIs and web services.

4. Why do we use head() after loading data?

Ans: The .head() method is used after loading data to get a quick overview of the dataset's structure, content, and to verify it loaded correctly.

5. What problems can occur while loading data?

Ans: Data loading issues commonly involve data corruption, technical limitations, and formatting errors.

Date: 8th March 2026

Topic: **Pandhas DataFrame**

Technology: Data Science

1. What is a Pandas DataFrame?

Ans: A Pandas DataFrame is a two-dimensional, size-mutable tabular data structure with labeled axes (rows and columns) in Python.

2. Difference between Series and DataFrame?

Ans: **DataFrame vs Series**

Series	DataFrame
One- dimensional	Two- dimensional
Series elements must be homogenous.	Can be heterogeneous.
Immutable(size cannot be changed).	Mutable(size can be changeable).
Element wise computations.	Column wise computations.

Functionality is less.	Functionality is more.
Alignment not supported.	Alignment is supported.

3. How do you create a DataFrame?

Ans: A DataFrame is a two-dimensional, tabular data structure, similar to a spreadsheet or SQL table, commonly used in libraries like Python's [pandas](#) and R. In Python, you use the `pd.DataFrame()` constructor from the pandas library to create one.

4. How do you access rows and columns in DataFrame?

Ans: Accessing rows and columns in a pandas DataFrame is primarily done using bracket notation (`[]`) and the explicit indexing methods: `.loc[]` (label-based) and `.iloc[]` (position-based).

5. What are the advantages of using DataFrame?

Ans: DataFrames provide a powerful, flexible, and organized 2D tabular data structure, making them superior for data manipulation, cleaning, and analysis compared to spreadsheets or simple arrays. Key advantages include handling heterogeneous data types, efficient handling of large datasets, intuitive row/column indexing, robust handling of missing values, and seamless integration with data science libraries.

Date: 8th March 2026

Topic: **Pandas Series**

Technology: Data Science

1. What is a Pandas Series?

Ans: A Pandas Series is a one-dimensional labeled array capable of holding data of any type (integers, strings, floats, Python objects, etc.).

2. Difference between Series and DataFrame?

Ans: Dimensionality: Series is 1D (rows only), DataFrame is 2D (rows and columns).

- Data Types: Series elements must be homogeneous (same type), while DataFrame columns can have mixed types.
- Structure:
A DataFrame is essentially a container for multiple Series objects.
- Mutability: Series size is generally immutable, while DataFrame size is mutable (can add/drop columns).

3. How do you create Series from dictionary?

Ans: To create a pandas Series from a dictionary, we use the `pandas.Series()` constructor, passing the dictionary as the main argument. The keys of the dictionary automatically become the index of the Series, and the values of the dictionary become the data.

Steps and Examples

1. Import the pandas library: By convention, it is imported with the alias pd.
2. Define your dictionary: The keys and values of the dictionary will be used for the Series.
3. Create the Series: Pass the dictionary to pd.Series().

4. How is indexing useful in Series?

Ans: In pandas Series, indexing is a critical feature that provides labels for each element in the data, serving as a powerful tool for data retrieval, alignment, and manipulation.

5. Can Series perform mathematical operations?

Ans: Yes, Pandans Series objects can perform mathematical operations.

Regards

Ashka Singh

----- Forwarded message -----

From: **ashka singh** <sgsinghashka@gmail.com>

Date: Tue, 10 Mar 2026 at 16:42

Subject: Fwd: TA || Ashka || TM- B141

To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Please be apprised that I am submitting the 28th Feb 26 assignment today, as the file was not available on the Top mentor LMS.

Regarding the topics of Classes and Objects, Exception Handling (try, except), File Handling (read, write), Object-Oriented Programming Basics, and Working with Built-in Modules, kindly find the requested details below.

Date: 28th Feb 2026

Topic: Classes and Objects

Technology: Python

1. What is a class?

Ans: A class in Python is a user-defined blueprint or template for creating objects.

2. What is an object?

Ans: In Python, an object is the fundamental abstraction for data, encapsulating both data (attributes or state) and the functions that operate on that data (methods or behavior).

3. What does ****init**** method do?

Ans: The **__init__** method in Python is a special "dunder" (double underscore) method that acts as a class constructor, automatically running when a new object instance is created. It initializes the object's attributes, sets the initial state, and assigns values to properties.

4. What is self in Python?

Ans: In Python, self is a convention-based name for the first parameter of an instance method within a class.

5. Can we create multiple objects from one class?

Ans: Yes, you can create multiple objects from a single class. This is a fundamental concept of object-oriented programming (OOP), where a class acts as a blueprint or template, and the objects are individual instances built from that blueprint.

Date: 28th Feb 2026

Topic: Exception Handling (try, except)

Technology: Python

1. What is exception handling?

Ans: Exception handling in Python is a mechanism used to manage runtime errors, known as exceptions, so that a program does not terminate abruptly.

2. What is purpose of try block?

Ans: The purpose of the try block in Python is to enclose code that might cause an error (an exception) during execution.

3. What does finally do?

Ans: In Python, the finally block is used within a try...except...finally statement to define a block of code that is guaranteed to execute under all circumstances, whether an exception occurred in the try block or not.

4. How do you handle specific errors?

Ans: In Python, you handle specific errors (exceptions) using try and except blocks, which allow our program to manage unexpected situations gracefully without crashing.

5. What happens if exception is not handled?

Ans: If an exception is not handled in Python, the program will terminate abruptly and stop executing.

Date: 28th Feb 2026

Topic: File Handling (read, write)

Technology: Python

1. What is file handling in Python?

Ans: File handling in Python is the process of managing the data flow between a program and files stored permanently on a computer's disk.

2. What is difference between r and w mode?

Ans: The 'r' mode is for reading existing files and raises an error if the file is not found, while the 'w' mode is for writing (and creating) files, and it overwrites the file's content if it already exists

3. Why do we use with open()?

Ans: The with open() statement is the standard and safest way to handle file operations in Python because it uses a context manager to ensure the file is automatically closed, even if errors occur.

4. How to read file line by line?

Ans: The most "Pythonic" and memory-efficient way to read a file line by line is to iterate over the file object within a with statement. This method ensures the file is automatically closed, even if errors occur, and only loads one line into memory at a time, making it suitable for large files.

5. What happens if file is not closed ?

Ans: Failing to close a file in Python can lead to data loss or corruption, resource exhaustion, and file access conflicts.

Date: 28th Feb 2026

Topic: Object-Oriented Programming Basics

Technology: Python

1. What is OOP?

Ans: Object-Oriented Programming (OOP) in Python is a programming paradigm that organizes code around objects rather than functions or logic.

2. What is difference between class and object?

Ans: A class is a blueprint or template that defines the common structure and behavior (attributes and methods) for a category of real-world entities. An object, on the other hand, is a concrete, individual instance created from that class, possessing actual data and existing in memory.

3. What is constructor?

Ans: A constructor is a special method used to initialize a new object's attributes and state when an instance of a class is created. The primary method used for this purpose is called `__init__()`.

4. What are OOP features?

Ans: Python is a multi-paradigm language that fully supports the principles and features of Object-Oriented Programming (OOP). The primary features are classes, objects, inheritance, encapsulation, polymorphism, and abstraction.

5. Why is OOP useful?

Ans: It provides a powerful and flexible way to structure code, making it more modular, reusable, and maintainable as programs grow in size and complexity.

Date: 28th Feb 2026

Topic: Working with Built-in Modules

Technology: Python

1. What are built-in modules?

Ans: Python's built-in modules are pre-installed libraries that come bundled with the standard Python installation.

2. How do you use math module?

Ans: To use the math module in Python, we must first import it using the import statement, and then access its functions and constants using dot notation (e.g., `math.sqrt()`). The module is part of Python's standard library and does not require separate installation.

3. What does random module do?

Ans: The Python random module is a built-in library used for generating pseudo-random numbers and performing random operations on sequences.

4. How to get current date in Python?

Ans: To get the current date in Python, you can use the built-in datetime module. The most straightforward method is using `date.today()` or `datetime.now().date()`.

5. What is sys module used for?

Ans: The sys module in Python provides access to variables and functions that interact with the Python interpreter and its runtime environment.

Regards

Ashka Singh

----- Forwarded message -----

From: **ashka singh** <sgsinghashka@gmail.com>

Date: Sat, 28 Feb 2026 at 01:40

Subject: Fwd: TA || Ashka || TM- B141

To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topics of List and list method, string, string method, sets, tuples, dictionaries, installing Python and IDE setup, keywords, identifiers, list comprehension, modules, and import statement. Kindly find the requested details below.

Date: 21st Feb 2026

Topic: List and list methods

Technology: Python

1 What is a list in Python?

Ans: Lists are used to store multiple items in a single variable.

2. How do you add item to a list?

Ans: Use the append() method to add a single element to the end of the list, or the extend() method to add multiple elements.

3. What is difference between append() and insert()?

Ans: Append adds to the end of the list, while insert adds in front of a specified index.

4. How to remove item from list?

Ans: Lists in Python have various built-in methods to remove items, such as remove, pop, del, and clear methods.

Removing elements from a list can be done in various ways, depending on whether we want to remove based on the value of the element or the index.

5. Can a list store different data types?

Ans: In a list, we can store objects of any type. We can also mix objects of different types within the same list, although list elements often share the same type.

Date: 21st Feb 2026

Topic: Sets

Technology: Python

1. What is a set in Python?

Ans: A set is one of 4 built-in data types in Python used to store collections of data.

2. Why are duplicate values not allowed in set?

Ans: Sets do not allow duplicate values, so they are useful for removing repeats from a list. Sets are unordered and unindexed, which means we cannot access items by position

3. What is difference between list and set?

Ans: Lists are ordered and can contain duplicate items. Sets are unordered, and all items must be unique.

4. How do you add and remove items from set?

Ans: We can add elements to the set using `add()` and remove elements from the set using `remove()`.

5. What are union and intersection in sets?

Ans: The `union()` and `update()` methods join all items from both sets. The `intersection()` method keeps ONLY the duplicates.

Date: 21st Feb 2026

Topic: Strings and String Methods

Technology: Python

1. What is a string in Python?

Ans: In Python, a string is a sequence of characters. For example, "hello" is a string containing a sequence of characters 'h', 'e', 'l', 'l', and 'o'.

2. What is string indexing?

Ans: Indexing allows us to access individual characters in a string directly by using a numeric value. String indexing is zero-based: the first character in the string has index 0, the next is 1, and so on.

3. What is string slicing?

Ans: String slicing in Python is a way to get specific parts of a string by using start, end and step values. It's especially useful for text manipulation and data parsing.

```
s = "Hello, Python!"
```

```
print (s[0:5])
```

Output

Hello

4. Name some common string methods.

Ans:

Method	Description
strip()	Returns a trimmed version of the string
swapcase()	Swaps cases, lower case becomes upper case and vice versa
title()	Converts the first character of each word to upper case
translate()	Returns a translated string

5. What does replace() do?

Ans: Replaces part of a text string, based on the number of characters we specify, with a different text string.

Date: 21st Feb 2026

Topic: Tuples

Technology: Python

1. What is a tuple in Python?

Ans: A Python tuple is a collection of ordered and immutable elements.

2. What does immutable mean?

Ans: The value of an object cannot be changed over time; then it is known as immutable.

3. What is difference between list and tuple?

Ans: Tuples are immutable objects, and lists are mutable objects. Once defined, tuples have a fixed length, and lists have a dynamic length.

4. How do you create single item tuple?

Ans: To create a tuple with a single item, we need to place the item followed by a comma.

5. Can we modify tuple values?

Ans: Once a tuple is created, we cannot change its values .

Date: 22nd Feb 2026

Topic: Dictionaries

Technology: Python

1. What is a dictionary in Python?

Ans: In Python, a dictionary (or dict) is a built-in data type for storing data in key-value pairs.

2. How do you access a value in dictionary?

Ans: Values in a Python dictionary can be accessed by placing the key within square brackets next to the dictionary.

Example:

```
a = {"Geeks": 3, "for": 2, "geeks": 1}
```

#Access the value associated with "geeks"

x = a["geeks"]

```
print(x)
```

Output

1

3. How to add new key–value pair?

Ans: Using square bracket notation. This method directly adds or updates the key-value pair in the dictionary.

4. What is difference between keys() and values()?

Ans: Keys must be unique and immutable. Values can be duplicated and of any data type.

5. Can dictionary keys be duplicate?

Ans: Duplicate keys are not allowed, but values can be duplicated.

Date: 22nd Feb 2026

Topic: Installing Python and IDE Setup

Technology: Python

1. What is the purpose of installing Python?

Ans: Python is a high-level, interpreted, and general-purpose dynamic programming language that focuses on code readability. It generally has smaller programs than Java and C.

2. What is an IDE and why is it useful?

Ans: (IDE) is a software application that helps programmers develop software code efficiently. It increases developer productivity by combining capabilities such as software editing, building, testing, and packaging in an easy-to-use application.

3. What happens if Python is not added to PATH?

Ans: We may be getting a message saying that the term Python isn't recognized, or we may end up with the wrong version of Python running.

4. What is the difference between IDLE and other IDEs?

Ans: IDLE is a simple Python-specific IDE for beginners, while IDEs are versatile development environments that support various programming languages and offer advanced features.

5. How can you check if Python is installed properly?

Ans: Press the Windows key or click on the Start button to open the Start menu. Type "python". If Python is installed, it should show up as the best match.

Date: 22nd Feb 2026

Topic: Keywords and Identifiers

Technology: Python

1. What are Python keywords?

Ans: Python keywords are special reserved words that have specific meanings and purposes and cannot be used for anything but those specific purposes.

2. What is an identifier?

Ans: An identifier may be a word, number, letter, symbol, or any combination of those that identifies (i.e., labels the identity of) either a unique object or a unique class of objects. identifiers (IDs) are symbols having the function to identify an entity or a data-set.

3. What are the rules for naming identifiers?

- Ans: A valid identifier can have letters (both uppercase and lowercase letters), digits, and underscores.
- The first letter of an identifier should be either a letter or an underscore.
- Cannot use keywords like int, while, etc. as identifiers.

4. Why can't we use keywords as identifiers?

Ans:

- Predefined and reserved words with special meanings.
 - Used to define the syntax and structure of Python code.
 - Cannot be used as identifiers, variables, or function names.
 - Written in lowercase, except True and False.
- `iskeyword()` → checks if a string is a keyword.

5. What happens if two identifiers differ only in case?

Ans: In Python, identifiers (such as variable, function, and class names) are case-sensitive. If two identifiers differ only in case, the Python interpreter treats them as completely distinct and independent entities.

Date: 22nd Feb 2026

Topic: List Comprehension

Technology: Python

1. What is list comprehension?

Ans: List comprehension offers a shorter syntax when we want to create a new list based on the values of an existing list.

2. How is it different from normal loop?

Ans: List comprehensions are often not only more readable but also faster than using "for loops."

3. Can we use condition in list comprehension?

Ans: We can use list comprehension with if statements to filter elements that meet certain conditions. This example selects only even numbers from the list of numbers.

4. Is list comprehension faster than loop?

Ans: List comprehensions are faster than for loops.

5. Where is list comprehension useful?

Ans: List comprehensions provide a more concise way to create lists in situations where map() and filter() and/or nested loops would currently be used .

Date: 22nd Feb 2026

Topic: Modules and Import Statement

Technology: Python

1. What is a module in Python?

Ans: A module is a file containing Python definitions and statements.

2. Why do we use modules?

Ans: They help in organizing code, making it more maintainable and scalable.

3. What is difference between import and from import?

Ans:

import <module> statement	from <module> import statement
It imports entire module.	It imports single, multiple or all objects from a module.

To access one of the functions, we have to specify the name of the module and the name of the function, separated by a dot. This format is called dot notation. The syntax is: <module-name>.<function-name>()	To access functions, there is no need to prefix module's name to imported item name. The syntax is: <function-name>
Imports all its items in a new namespace with the same name as of the module.	Imports specified items from the module into the current namespace.
This approach does not cause any problems.	This approach can lead to namespace pollution and name clashes if multiple modules import items with the same name.
For example: <pre>import math print(math.pi) print(math.sqrt(25))</pre>	For example: <pre>from math import pi, sqrt print(pi) print(sqrt(25))</pre>

4. What is alias in import statement?

Ans: In Python, an alias in a `import` statement is an alternative name assigned to a module, class, or function using the `as` keyword. This allows us to refer to the imported object by a shorter or more convenient identifier within your code.

Import Type	Syntax	Usage in Code
Module Alias	<code>import math as m</code>	<code>m.sqrt(16)</code>
Member Alias	<code>from datetime import datetime as dt</code>	<code>now = dt.now()</code>
Submodule Alias	<code>import matplotlib.pyplot as plt</code>	<code>plt.plot(x, y)</code>

5. Name some built-in Python modules.

Ans: Python Module Index

<code>cmath</code>	Mathematical functions for complex numbers.
<code>cmd</code>	Build line-oriented command interpreters.

code	Facilities to implement read-eval-print loops.
codecs	Encode and decode data and streams.
codeop	Compile (possibly incomplete) Python code.

Thanks & Regards

Ashka Singh

----- Forwarded message -----

From: ashka singh <sgsinghashka@gmail.com>

Date: Fri, 20 Feb 2026 at 22:10

Subject: Fwd: TA || Ashka || TM- B141

To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topics of break, continue, pass, for loop, while, function arguments, functions, and lambda functions, kindly find the requested details below.

Date: 14th Feb 2026

Topic: Break, continue, pass

Technology: Python

1. What is the purpose of break statement?

Ans: The `break` statement in Python is used to prematurely terminate a `for` or `while` loop. When encountered, it immediately stops the current loop iteration and exits the loop, transferring control to the next line of code after the loop. It is primarily used to optimize code or exit when a specific condition is met

2. How is continue different from break?

Ans: Break terminates the entire loop, while continue skips only the current iteration and proceeds to the next one.

3. When do we use pass?

Ans: When a statement is syntactically required, but we do not want any code or command to execute.

4. Can break be used outside a loop?

Ans: No, the `break` statement in Python can only be used inside a `for` or `while` loop. If you attempt to use it outside of a loop, it will raise a `SyntaxError: 'break' outside loop`.

5. What happens after continue is executed?

Ans: In a for loop: Control jumps to the loop's iterator to process the next item in the sequence. If there are no more items to process, the loop terminates.

- In a while loop: Control immediately jumps back to the top of the loop to re-evaluate the loop's condition. If the condition is still True, the next iteration begins; if False, the loop terminates.

Date: 14th Feb 2026

Topic: For Loop

Technology: Python

1. What is a for loop?

Ans: A control flow statement used for iterating over a sequence (such as a [list](#), [tuple](#), [string](#), [dictionary](#), or [range](#) object) or other iterable objects.

2. How does range() work?

Ans: The range() function returns a sequence of numbers, starting from 0 by default, and increments by 1 (by default), and stops before a specified number.

3. Can we loop through a string?

Ans: Yes, we can easily loop through a string because strings are an ordered sequence of characters, making them inherently iterable.

4. What is the difference between for and while loop?

Ans: [For loops](#) are best for iterating a specific number of times (e.g., over a range or list), while [while loops](#) repeat code as long as a condition remains true.

5. How do you stop a loop early?

Ans: In Python using the break statement. This statement immediately terminates the loop (both for and while loops) and the program continues execution at the next line of code after the loop block.

Date: 14th Feb 2026

Topic: While Loop

Technology: Python

1. What is a while loop?

Ans: A control flow statement that repeatedly executes a block of code as long as a specified condition remains True.

2. When do we use while instead of for?

Ans: A while loop, when the number of iterations is unknown beforehand, and the loop should continue until a specific condition becomes false. Use a for loop when the number of iterations is fixed or known in advance, typically when iterating over a sequence.

3. What causes an infinite loop?

Ans: The program might be stuck in an endless loop because the terminating condition is never satisfied. This could be due to logic errors, incorrect variable updates, or missing increment/decrement statements in your loop.

4. How can you stop a while loop?

Ans: Ensuring its governing condition becomes false, by using a break statement to exit immediately, or by returning from a function containing the loop.

5. Can while loop run without condition?

Ans: No, a while loop cannot run without a condition. The syntax requires a boolean expression (or an expression that evaluates to truthy or falsy) that determines whether the loop continues or terminates. However, we can create a loop that runs indefinitely by providing a condition that is always true, such as while True:

Date: 15th Feb 2026

Topic: Function Arguments

Technology: Python

1. What are function arguments?

Ans: The actual values or variables passed as input to a function when it is called (invoked).

2. What is default argument?

Ans: A pre-assigned value for a function parameter that the compiler or interpreter uses if no value is provided during the function call.

3. What is keyword argument?

Ans: An argument is the actual value or object passed into a function when it is called.

python

```
def greet(name, greeting="Hello")
```

```
    """Prints a greeting message."""
```

```
    print(f'{greeting}, {name}!')
```

```
    greet("Alice")
```

```
    greet("Bob", "Hi")
```

```
    greet(name="Charlie", greeting="Good morning")
```

Output:

Hello, Alice!

Hi, Bob!

Good morning, Charlie!

4. Can we mix default and keyword arguments?

Ans: Yes, we can mix default and keyword arguments in Python, but there is one strict rule: all positional arguments must come before any keyword arguments in the function call.

5. What happens if default value is not provided?

Ans: If a default value is not provided for a parameter, variable, or database field, the system typically assigns a default "absence" value, such as undefined, null, or 0.

Date: 15th Feb 2026

Topic: Functions

Technology: Python

1. What is a function in Python? Ans: A function in Python is a named, reusable block of code that performs a specific task when called.

2. Why are functions important?

Ans: Functions are essential in Python because they allow developers to organize code into reusable, logical blocks, which makes programs easier to read, debug, and maintain.

3. What is difference between print and return?

Ans: **Return** exits a function, whereas **print** simply displays output.

4. What are parameters in function?

Ans: Function parameters are variables defined within the parentheses in a function's definition that act as placeholders for the actual values (arguments) that will be passed when the function is called.

5. Can a function run without parameters?

Ans: Yes, a function in Python can be defined and called without any parameters.

```
def greet():
```

```
    """Prints a greeting message."""
```

```
    print("Hello, how are you?")
```

```
greet()
```

Output:

Hello, how are you?

Date: 15th Feb 2026

Topic: Lambda Functions

Technology: Python

1. What is a lambda function?

Ans: A lambda function is a small, anonymous function defined without a name, typically used for short-term, one-off operations.

A lambda function that adds 10 to an input number

```
add_ten = lambda x: x + 10
```

```
print(add_ten(5)) # Output: 15
```

2. How is lambda different from normal function?

Ans: A lambda is an anonymous, single-expression function designed for short, one-time use, while a normal function is named, can contain multiple statements and expressions, and is designed for reusability and complex logic.

3. Where are lambda functions used?

Ans: Primarily used for short, one-time operations as arguments to higher-order functions.

4. Can lambda have multiple arguments?

Ans: Yes, a lambda function can have multiple arguments. Example `multiply = lambda x, y: x * y`

```
print(multiply(2, 3)) # Output: 6
```

5. Why are lambda functions called anonymous?

Ans: Lambda functions are called "anonymous" because they are defined without a name, unlike standard functions created with the def keyword in Python.

Thanks & Regards

Ashka Singh

----- Forwarded message -----

From: ashka singh <sgsinghashka@gmail.com>

Date: Thu, 19 Feb 2026 at 17:24

Subject: Fwd: TA || Ashka || TM- B141

To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topics of data type, output statement, Python syntax rules, taking user input, types of conversion, and variables, kindly find the requested details below.

Date: 7th Feb 2026

Topic: Data Type

Technology: Python

1. What are data types in Python?

Ans: 1 The following are standard or built-in data types in Python:

- Numeric: [int](#), [float](#), [complex](#)
- Sequence Type: [string](#), [list](#), [tuple](#)
- Mapping Type: [dict](#)
- Boolean: [bool](#)
- Set Type: [set](#), [frozenset](#)
- Binary Types: [bytes](#), [bytearray](#), [memoryview](#)



2. What is the difference between int and float?

Ans :

Feature	Integer	Float
Definition	Whole numbers without decimal points	Numbers with decimal points
Declaration	e.g., $x = 5$	e.g., $y = 3.14$
Precision	Infinite precision	Limited precision
Operations	Supports arithmetic operations	Supports arithmetic operations

Division	Integer division returns an integer (floor)	Division returns a float
Memory	Typically occupies less memory	Typically occupies more memory
Example	"x = 5"	"y = 3.14"
Range	Limited by system memory and resources	Limited by floating-point representation
Representation	Stored as binary numbers without fractions	Stored in IEEE 754 floating-point format
Accuracy	Accurate for counting and discrete values	May have rounding errors and representation issues
Conversion	Can be explicitly converted to a float	Can be explicitly converted to an integer; rounding may occur

Use Cases	Ideal for counting, indexing, and whole numbers	Ideal for continuous data, measurements, and calculations
Syntax	Written without a decimal point	May include a decimal point and/or exponent

3. How is string data written in Python?

Ans: In Python, string data is written as a sequence of characters enclosed within quotation marks.

Method	Example	Use Case
Single Quotes	name = 'Alice'	Standard single-line strings.
Double Quotes	city = "New York"	Standard single-line strings; useful when the string itself contains a single quote (e.g., "It's a great day").
Triple Quotes	paragraph = """This string spans multiple lines and preserves newlines."""	Multiline strings and docstrings

4. What are Boolean values used for?

Ans: Boolean data is fundamental in programming. It represents two possible values, which are either true or false. It is used to perform logical operations and control conditional statements.

5. How can you check the type of a variable?

Ans: We can check the type of a variable in Python using the built-in type() and isinstance() functions.

Date: 7th Feb 2026

Topic: Output Statement

Technology: Python

1. What is the purpose of print() in Python?

Ans: The print() function/ statement in Python is a **built-in function** used to output information to the console/ standard output device. It helps programmers display data, debug code, or present results during program execution.

2. How can you print multiple values in one line?

Ans: In Python, you can print multiple values on a single line using several methods, including separating them with commas in a single print() call, using f-strings, or modifying the end parameter of the print() function.

3. What does \n do in print()?

Ans: In print(), the \n character acts as a newline escape sequence, which moves the output cursor to the beginning of the next line.

4. Can print() display numbers and text together?

Ans: Yes, the print() function in Python can easily display text (strings) and numbers together. There are several methods to achieve this, with the most common ones involving separating the items with commas or using string formatting

5. What happens if print() is not used in a program?

Ans: The program will run its logic internally but will not display any results or information to the user on the console.

Date: 7th Feb 2026

Topic: Python Syntax Rules

Technology: Python

1. What is syntax in Python?

Ans: Syntax refers to the specific rules and structure for writing valid code in Python, including the proper use of the or keyword.

2. Why is indentation important in Python?

Ans: It is a mandatory part of the syntax used to define the structure and scope of code blocks.

3. What happens if indentation is wrong?

Ans:

- Python enforces indentation to define the scope of loops, functions, conditionals, and other blocks of code.
- If your indentation is inconsistent or misplaced, Python will throw an error before it even runs the program.
- That's why this is considered a syntax error (detected before execution).

4. Why is Python called case-sensitive?

Ans: Python differentiates between uppercase and lowercase characters, making it a case-sensitive programming language. This distinction means that Python treats identifiers such as variable, Variable, and VARIABLE as entirely separate entities.

5. What are the rules for naming variables?

Ans: Variable names should be meaningful so that they are easy to remember too.

☞ Variable name can be of any size.

☞ Variable name can have characters a-z, A-Z, 0-9, and Underscore(_).

☞ Variable name should begin with a letter or an underscore.

☞ Variable names are case sensitive. e.g., 'total' and 'TOTAL' are not the same.

☞ Always remember that a variable name should be a keyword, such as else, print, if, etc.

Date: 7th Feb 2026

Topic: Taking User Input

Technology: Python

1. What is the purpose of input() in Python?

Ans: To allow a program to interact with the user by accepting data from the standard input device, usually the keyboard, during program execution.

2. Why do we use int() with input()?

Ans: In Python, we use int() with input() because the input() function always returns the user's entry as a string (even if it looks like a number), and the int() function is needed to convert that string into an integer data type so that mathematical operations and comparisons can be performed correctly.

3. What type does input() return by default?

Ans: The input() function always returns a string.

4. How can you take multiple inputs from the user?

Ans: Using split() method - Using list() function - Using map() function - Using raw_input() function

5. What happens if the user enters text, but int() is used?

Ans: The Python program will stop its execution and raise a ValueError.

Date: 7th Feb 2026

Topic: Type Conversion

Technology: Python

1. What is type conversion in Python?

Ans: the process of changing a value from one data type to another.

2. What is the difference between implicit and explicit conversion?

Ans:

Implicit Type Conversion	Explicit Type Conversion
An implicit type conversion is automatically performed by the compiler when differing data types are intermixed in an expression.	An explicit type conversion is user-defined conversion that forces an expression to be of specific type.

An implicit type conversion is performed without programmer's intervention.	An explicit type conversion is specified explicitly by the programmer.
Example: <pre>a, b = 5, 25.5 c = a + b</pre>	Example: <pre>a, b = 5, 25.5 c = int(a + b)</pre>

4. Can we convert the string "abc" to an int? What happens?

Ans: No, we cannot convert the string "abc" to an int using standard built-in functions in most programming languages because it does not represent a valid numerical value.

Attempting to do so will result in a runtime error or exception.

5. When does Python perform automatic conversion?

Ans: Python performs automatic type conversion primarily during operations involving different but compatible numeric data types.

Date: 7th Feb 2026

Topic: Variables

Technology: Python

1. What is a variable in Python?

Ans: A variable is a symbolic name that acts as a reference or pointer to an object stored in memory. Variables are used as containers for storing data values, such as numbers, strings, and lists, making it easy to access and manipulate that data throughout a program.

2. How do you assign a value to a variable?

Ans: Python Variables: Unlike other languages, we can easily assign values to multiple variables in python easily.

Assigning multiple values in single line

```
a,b,c="geeks","for","geeks"
```

```
print(a+b+c)
```

Output

Geeksforgeeks

4. Can a variable value change during program execution?

Ans: Yes, the fundamental characteristic of a variable in programming is that its value can change during the execution of a program.

5. What happens if two variables have the same name but different cases?

Ans: In Python, variables with the same name but different cases are treated as completely different variables.

Date: 8th Feb 2026

Topic: Conditional Statements

Technology: Python

1. What is the purpose of conditional statements?

Ans: They allow your program to evaluate conditions and execute specific code based on the outcomes only when certain conditions are true or false.

2. What is the difference between if and if-else?

Ans: if Statement: Checks a condition. If True, code runs. If False, nothing happens, and the program proceeds.

- **if-elseStatement**: Checks a condition. If True, the if block runs. If False, the else block runs.

3. When do we use elif?

Ans: When we have multiple, mutually exclusive conditions that we want to check in a specific, sequential order.

4. What is a nested if statement?

Ans: A nested if statement is an if structure placed inside another if or else block. It enables checking multiple, dependent conditions in a sequence, where the inner condition is only evaluated if the outer condition is true. This structure allows for complex decision-making, though deep nesting can make code difficult to read.

5. Can we use multiple elif blocks?

Ans: Yes, we can use multiple elif blocks in an if statement chain. This structure, often called a chained conditional, allows us to check multiple conditions sequentially, and the first condition that evaluates to True will have its corresponding code block executed, while all subsequent elif and else blocks will be skipped.

Date: 8th Feb 2026

Topic: Nested Conditions

Technology: Python

1. What is a nested condition?

Ans: A nested condition, or nested conditional, is a programming structure where an if or else statement is placed inside another if or else statement.

2. Why do we use nested if statements?

Ans: A nested if statement is used to evaluate multiple, dependent conditions by placing an if or if-else structure inside another if block. It enables multi-level decision-making—where the inner condition only executes if the outer condition is true—making it ideal for complex logic, such as grading systems, login validation, and eligibility checks.

3. What happens if the outer condition is False?

Ans: The entire inner block is skipped, and control passes to the corresponding else block or the next statement in the program. The inner code is never evaluated, preventing any actions nested within that outer condition from running.

4. Can nested conditions have an else block?

Ans: Yes, nested conditions can absolutely have else blocks.

5. How do nested conditions help in real-world programs?

Ans: Real-world programming for handling complex, hierarchical decision-making, such as validating user credentials, processing tiered discounts, or implementing multi-stage security protocols.

Date: 8th Feb 2026

Topic: Operators

Technology: Python

1. What are operators in Python?

Ans: Operators are special symbols or keywords used to perform operations on values and variables (called operands), such as mathematical calculations, comparisons, and logical evaluations.

2. What is difference between = and == ?

Ans: Purpose: = assigns a value; == compares values.

- Result: = returns the assigned value; == returns a boolean (true/false) or a similar indicator (e.g., 1 or 0).
- Usage: a = 10 sets variable a to 10; a == 10 checks if a is 10.

3. What are logical operators used for?

Ans: To combine conditional statements and perform logical operations on Boolean values (True or False).

4. What does % operator do?

Ans: An operator in Python is a special symbol or keyword that tells the interpreter to perform a specific operation on values or variables

5. Explain assignment operators with an example?

Ans: Assignment operators in Python are used to assign values to variables. They can be simple, like the equals sign (=).

Example

```
x = 5
y = "hello"
print(x)
print(y)
```

Output:

```
5
hello
```

Thanks & Regards
Ashka Singh

----- Forwarded message -----

From: ashka singh <sgsinghashka@gmail.com>

Date: Fri, 6 Feb 2026 at 16:42

Subject: TA || Ashka || TM- B141

To: <dhiraj.aiworkspace@gmail.com>

Dear Sir,

Regarding the topics of the Case study on Quick company (Blinkit), introduction of Python, and introduction to data science, kindly find the requested details below.

Date: 31st January 2026

Topic: Case study on Blinkit

Technology: Data Science

Ques 1: What is the role of the data analyst at Blinkit?

Ans: A Data Analyst at Blinkit is driving operational efficiency and business growth by analyzing, visualizing, and interpreting complex data. They collaborate with cross-functional teams to optimize workflows, create data-driven reports, and identify strategic opportunities using SQL, Python/R, and visualization tools like Tableau.

1. Operations & Performance

- **Order Tracking: Analyzing hourly order volume and revenue to ensure sales targets are met.**
- **Delivery Monitoring: Breaking down the "10-minute" window to find bottlenecks—whether it's slow packing in the dark store or rider delays.**

2. Product & Inventory

- **Sales Ranking: Identifying high-velocity products (Top SKUs) to ensure they never go out of stock.**
- **Wastage Control: Tracking expiry dates and slow-moving items to minimize losses, especially in perishables like milk and vegetables.**

3. Reporting & Automation

- **Dashboards: Building real-time visualizations in Tableau or Power BI so managers can see live store health.**
- **Actionable Insights: Providing "Root Cause Analysis" (RCA) when KPIs dip—moving beyond just "what" happened to "why" it happened.**

Ques 2: How does data help Blinkit improve delivery speed?

Ans. . Data helps Blinkit hit the 10-minute mark by optimizing three critical stages:

1. Predicting Demand (Pre-Order)

- Hyperlocal Inventory: Data predicts what a specific neighborhood buys (e.g., more milk in suburbs, more snacks in tech hubs), so items are already in the nearest warehouse.
- Smart Layout: High-demand items are moved to the "Golden Zone" (front of the store) to reduce "picking" time to under 2 minutes.

2. Reducing Friction (In-Store)

- Bottleneck Detection: If a store takes 3 minutes to pack instead of 2, analysts identify if it's due to understaffing or poor shelf organization.
- Auto-Replenishment: Real-time data triggers stock refills before a product runs out, preventing "out-of-stock" cancellations.

3. Precision Logistics (Post-Order)

- Rider Assignment: Algorithms assign riders based on their proximity and predicted availability, not just distance.
- Dynamic Routing: GPS data identifies the fastest shortcuts, avoiding traffic and road closures in real-time.

Ques 3: What type of data is collected by Blinkit?

Ans: Blinkit collects a massive variety of data points to synchronize its "10-minute" promise. This data can be grouped into four main buckets:

1. Customer Behavior Data

- Search & Intent: What consumers search for, what consumers add to their cart but don't buy, and how much time they spend on specific product pages.
- Order Patterns: Frequency of purchases, average basket size, and preference for certain time slots (e.g., morning milk vs. late-night snacks).
- Location Hubs: Precise GPS coordinates of customers' delivery addresses to map "demand hotspots."

2. Operational & Logistics Data

- Dark Store Metrics: Real-time inventory levels, picking time (how fast staff bag an order), and stock-out frequencies.
- Rider Telematics: Live GPS tracking of delivery partners, transit speeds, and idle times at storefronts.
- External Factors: Real-time weather data (rain/heat) and local traffic conditions, which immediately trigger "surge pricing" or "extended delivery" warnings.

3. Product & Brand Data

- SKU Performance: Sales velocity of every single item (e.g., Amul Milk vs. Mother Dairy) and its shelf-life/expiry tracking.
- Pricing & Discounts: Competitor price monitoring and the effectiveness of "Buy 1 Get 1" or flash deal banners.
- Category Health: Which categories are growing (e.g., the recent shift toward selling electronics and toys alongside groceries).

4. Technical & Personal Data

- Device Info: IP addresses, phone models, and app version to fix bugs and optimize the user interface.
- Communication logs: Chats between the customer and the delivery partner (usually anonymized or recorded for safety/quality).

Ques 4: How can demand prediction help Blinkit reduce losses?

Ans: Demand prediction is the key to Blinkit’s efficiency, ensuring that the right products are in the right place at the exact time they are needed. By utilizing hyperlocal data and real-time monitoring, the platform significantly cuts down on various forms of financial loss.

 How Demand Prediction Saves Money

- **Minimizing Spoilage:** AI forecasts demand for perishables like milk and vegetables at a neighborhood level. If sales for items like tomatoes slow down, the system triggers automatic flash discounts to clear inventory before it expires.
- **Cutting Holding Costs:** Analysts identify "slow movers" and remove them to free up space in small, expensive urban warehouses. Using just-in-time replenishment, stores receive daily stock updates based on actual sales, keeping capital fluid.
- **Preventing Lost Revenue:** Early warning systems alert managers when popular items are nearly sold out. If a stock-out does occur, the app uses substitute logic to suggest similar items, ensuring the customer doesn't leave the platform for a competitor.

Risk Area	Data Solution	Financial Benefit
Fresh Produce	Hyperlocal Forecasting	Lower Wastage
Warehouse Space	SKU Optimization	Lower Rent/Unit
Inventory	Just-In-Time Stocking	Better Cash Flow
Stock-Outs	Early Warning Alerts	Higher Revenue

Question 5: What challenges can data scientists face in quick-commerce companies?

Answer: In quick commerce, the margin for error is nearly zero. Data scientists at companies like Blinkit and Zepto face "high-stakes" challenges where a minor model glitch can physically disrupt a city's supply chain in minutes.

 1. High-Velocity Data Latency

In quick commerce, data is generated every second (rider GPS, stock levels, order updates).

- **The Challenge:** Traditional data warehouses often have "sync latency." If a data scientist builds a model on 10-minute-old data, the "Picking" status of an order or the location of a rider might already be obsolete.
- **Impact:** This makes real-time dynamic routing and automated dispatching incredibly difficult to keep accurate.

2. The "Hyperlocal" Cold Start Problem

Blinkit operates through hundreds of "dark stores," and each one behaves like its own mini-market.

- The Challenge: When a new dark store opens in a new neighborhood, there is zero historical data for that specific area.
- Impact: Predicting demand for perishables becomes a guessing game. Data scientists must use "proxy data" from similar neighborhoods, which often leads to initial overstocking or stock-outs.

3. Extreme Sensitivity to External Factors

Quick commerce is highly volatile and reacts instantly to the "outside world."

- The Challenge: A sudden rainstorm, a local cricket match, or a viral social media trend can cause an instant 500% spike in specific orders (e.g., umbrellas or snacks).
- Impact: Standard forecasting models often fail to account for these non-linear spikes, leading to rider shortages or empty shelves.

4. Thin Margins vs. Complex Models

Quick commerce operates on incredibly thin profit margins.

- The Challenge: Building a highly complex "Deep Learning" model might improve accuracy by 1%, but the computational cost (server bills) might be higher than the profit gained from that 1%.
- Impact: Data scientists must prioritize "computationally cheap" and interpretable models over complex black-box ones to keep the Cost Per Order (CPO) low.

5. The "Expiry Blind Spot."

Managing fresh produce requires FEFO (First-Expired-First-Out) logic.

- The Challenge: Often, the system knows *how many* tomatoes are in a store, but it doesn't know the *exact age* of each individual batch in real-time due to manual entry errors at the warehouse.
- Impact: This makes it hard to create a "Dynamic Discounting" model that automatically lowers prices on items about to spoil.

Challenge Summary

Challenge	Impact
Data Sync	Inaccurate delivery ETAs.
No History	High wastage in new locations.
External Spikes	Rider shortages during rain.

Compute Cost	Direct hit to profit margins.
--------------	-------------------------------

Date: 31st January 2026

Topic: Introduction to Data Science

Technology: Data Science

Question 1. What is Data Science?

Ans: Data Science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract actionable knowledge and insights from structured and unstructured data.

It is essentially the "detective work" of the digital age—taking raw, messy information and turning it into a story that helps businesses and researchers make better decisions.

The Three Pillars of Data Science

Data Science sits at the intersection of three distinct areas:

- Computer Science: Using code (Python, R, SQL) to automate data processing and build models.
- Mathematics & Statistics: Applying probability and linear algebra to find patterns and ensure results aren't just coincidences.
- Domain Expertise: Understanding the specific "business" problem (e.g., healthcare, finance, or sports) to ask the right questions.

The Data Science Lifecycle

A typical project follows a standard flow:

1. Capture: Gathering raw data from databases, web scraping, or sensors.
2. Maintain: Cleaning the data (handling missing values or "junk" data).
3. Process: Exploring the data to find trends or anomalies.
4. Analyze: Performing "Predictive Analysis" or Machine Learning.
5. Communicate: Creating charts and reports to explain findings to non-technical people.

Real-World Examples

- Netflix: Uses data science to predict which movies you'll like based on your viewing history.
- Logistics: Companies like UPS use it to find the most fuel-efficient delivery routes.
- Healthcare: Predicting the spread of diseases or identifying tumors in X-rays through computer vision.

Ques 2: Why is Data Science important today?

Answer: In 2026, Data Science is essential because it acts as the "refinery" for the 400 million terabytes of data produced daily.

- Precision Decision-Making: It replaces "gut feelings" with evidence, allowing businesses to predict market shifts and customer needs with high accuracy.
- The Engine of AI: Modern AI and automation (like autonomous drones or smart assistants) only function because Data Science provides the cleaned data and optimized models they run on.

- **Real-Time Efficiency:** It allows for "living" systems, such as banks stopping fraud the millisecond it happens or logistics companies rerouting trucks instantly to avoid delays.
- **Global Impact:** It is the primary tool for solving complex human issues, from accelerating drug discovery to optimizing renewable energy grids for smart cities.



Ques 3: What type of data is used in Data Science?

Ans:



Ques 4: How is Data Science different from data analysis?

Ans: While both roles involve working with data, the primary difference lies in the timeframe and complexity of their goals.

Key Differences at a Glance

Feature	Data Analyst	Data Scientist
Primary Goal	Describing the past/present.	Predicting the future.
Tools	SQL, Excel, Tableau/PowerBI.	Python, R, Machine Learning, Spark.
Output	Reports, Dashboards, KPIs.	Predictive Models, Algorithms.

Skill Set	Statistics & Data Visualization.	Advanced Coding & Math.
-----------	----------------------------------	-------------------------

Ques 5. What problems can Data Science solve in real life?
Ans: In 2026, Data Science has evolved from a "corporate luxury" into a primary tool for solving complex global and daily challenges. It moves us from reacting to problems to predicting and preventing them. The most impactful real-life problems Data Science solves today:

Summary Table

Sector	Problem Solved	Data Science Benefit
Healthcare	Delayed diagnosis	Early detection via AI imaging
Finance	Identity theft	Real-time anomaly detection
Logistics	Inefficient delivery	Fuel-saving route optimization
Environment	Resource waste	Precision farming and energy grids

Date: 31st January 2026
Topic: Introduction to Structured Data, Unstructured Data, Semi-Structured Data
Technology: Data Science

Ques 1. What is structured data?

Ans: Structured data is information that is highly organized and formatted in a way that is easily searchable in relational databases. Think of it as data that fits perfectly into a spreadsheet or a SQL table.

Key Characteristics

- Quantitative: Usually consists of numbers and short strings (like dates or names).
- Fixed Schema: The data follows a strict model; you must define what the data is (e.g., "Zip Code" must be 5 digits) before you can save it.
- Relational: Data points are linked through keys. For example, a "Customer ID" in a sales table links to a "Customer Name" in a profile table.

Ques 2: Give examples of unstructured data.

Answer: Unstructured data is information that doesn't fit into a standard grid or database. It makes up roughly 80% of all data today.

Key Examples:

- Text: Emails, social media posts, PDFs, and chat logs.
- Audio: Phone call recordings, podcasts, and voice notes.
- Video: YouTube clips, surveillance footage, and movies.
- Images: Photos, medical X-rays, and satellite imagery.

The Challenge: Unlike a spreadsheet, a computer cannot "read" this data without using Artificial Intelligence or Machine Learning to find patterns

Ques 3: What makes semi-structured data different?

Ans: Semi-structured data is the "middle ground" of data. It doesn't live in rigid rows and columns, but it uses tags or keys to organize information.

What makes it different:

- Flexible Schema: Unlike a spreadsheet, every entry doesn't have to be the same. One user can have a "Twitter handle" while another doesn't, and the system won't crash.
- Self-Describing: